The Impact of Female Teachers on Female Students' Lifetime Well-Being

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Abstract

We document an enduring influence of female primary school teachers on their female students in rural America circa 1940. Using historical US Census data linked to IRS records, later censuses, and death records, we find that exposure to female teachers significantly improved girls' educational attainment—particularly among those with poorly-educated parents and in areas with limited labor market opportunities for women. These educational gains translated into delayed marriage and childbearing, marriage matches to better-educated husbands and higher household income. We also find that exposure to female teachers increased longevity of women relative to men.

Keyword: Female Teachers; Education and Gender Equality; Education and Mortality

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[*Kids*] *don't remember what you try to teach them. They remember what you are.* —Jim Hensen

I. Introduction

Do female students benefit from being taught by female teachers? The literature suggests that in some settings exposure to female teachers improves girls' and young women's educational outcomes (e.g., Carrell et al., 2010; Porter and Serra, 2010; Paredes, 2014; Lim and Meer, 2017, 2020), while in other settings it does not (e.g., Holmlund and Sund, 2010; Cho, 2012; Antecol et al., 2015; Schaede and Mankki, 2023). A close reading of these studies suggests that female teachers are most beneficial in environments where there are few female role models to demonstrate the value of education, as is often true in developing countries, and more broadly in STEM education.[1] In these contexts female teachers may help overcome the "powerful and sticky stereotypes about gender-specific skills and gender-specific roles," emphasized by Bertrand (2020).

In this paper, we examine the impact of female public school teachers in rural areas of the United States circa 1940—a time when women were significantly less likely than men to attend college or hold prominent professional roles. Well-educated women rarely occupied visible positions in business or politics, and female representation in leadership was minimal at both local and national levels.[2] In the rural communities we study, female primary school teachers often played a dual role—not only as educators but also as influential role models for their female students.

We focus on women born in the 1920s and early 1930s. These women were on the leading edge of what Goldin (2014) calls "the quiet revolution"—a revolution that ultimately resulted in profound changes in US society. They were more likely than their mothers to attend college and far more likely to build professional careers. Some went on to break longstanding gender barriers in science, law, and politics.[3]

The first contribution of our paper is to show that in our setting, exposure to a higher fraction of female primary school teachers led female students to achieve higher rates of school enrollment and grade completion during the critical transition from elementary to high school—specifically at ages 15 to 18. We estimate these impacts using models that

[1] Similar observations pertain concerning the benefits of matching the race/ethnicity of students and teachers. See, e.g., Redding (2019) for a systematic review, and Gershensen et al. (2021) for a carefully-designed recent analysis.

[2] No Fortune 500 company was led by a woman until 1972. In the 76th U.S. Congress (1939–1941), only eight of the 435 House seats and one of the 96 Senate seats were held by women—unsurprising given that women had gained the right to vote only two decades earlier.

[3] Examples include Elinor Ostrom (born 1933), the first woman to win the Nobel Prize in Economics, and Sandra Day O'Connor (born 1930) and Ruth Bader Ginsburg (born 1933), the first female justices to serve on the US Supreme Court.

compare boys and girls in the same county, census enumeration district, or household, and control for a rich set of family background and neighborhood characteristics (e.g., the employment rate of nearby young women), all with gender-specific impacts. We observe patterns that are consistent with the role model framework: the impact of female teachers is more pronounced for daughters of poorly-educated parents, and in regions where labor market opportunities for women are relatively scarce.

Our second distinctive contribution is an evaluation of *lifetime outcomes.* We find that women taught by female teachers were more likely to complete high school and attend college, experienced delayed marriage and childbearing, partnered with better-educated spouses, earned higher lifetime family income, and lived longer. Documenting these long-run effects is essential for assessing the broader benefits of female teachers. It also addresses a common concern: that short-run impacts of educational interventions (e.g., improvements in grades or test scores) may provide limited insight into their enduring consequences.[4]

We take the following four steps in developing our empirical evidence:

The first step is to use 1940 Census data to construct a variable that measures the gender composition of likely teachers for each child—the local fraction female among teachers (FFT). In 1940 Census enumerators completed "data sheets" residence-by-residence as they moved along streets within assigned "enumeration districts" (small geographic areas). We exploit this data structure to identify teachers who are close neighbors for each child—a strategy that is viable in rural areas, but not in larger cities, where the gender mix of a child's neighboring teachers is less informative about the FFT in the school attended by that child.[5]

Our second step is to estimate models of the relationship between the local FFT and contemporaneous schooling outcomes for white 15–18 year olds who lived with their parents at the time of the 1940 Census. We begin with "county models" that include county fixed effects, estimating the impacts of local differences in FFT within counties, on male and female students separately. We then estimate comparable "enumeration district (ED) models" that feature ED-level fixed effects. Since our FFT measure is at the ED level, in these models we can only estimate the relative effect of more female teachers on female students compared to male students, i.e., a differences-in-differences style

[4] A prominent example is provided by Heckman, Pinto, and Savelyev (2013), who estimate substantial long-term benefits of the Perry Preschool Program—including reduced criminal activity and increased employment—despite the program's short-lived effects on childhood cognitive test scores. Their findings highlight non-cognitive skill development, particularly enhanced academic motivation among girls.

[5] Our FFT variable is subject to measurement error. We use administrative data to quantify this error and apply a correction procedure to obtain error-adjusted parameter estimates. A detailed discussion of this approach is provided later in the paper (see Section IV.C).

estimate that reflects how differences in FFT affect the relative educational outcomes of girls and boys in the same EDs. Finally, we estimate "household models" that compare the outcomes of sisters relative to their brothers.

In all models we include a detailed set of control variables, including child and family characteristics, county characteristics drawn from historical sources, and neighborhood socioeconomic features (variables we construct by averaging the characteristics of families nearby a child's home). Importantly, all these variables are interacted with child gender, i.e., in our specifications all factors (not just FFT) are allowed to have differing impacts on the educational success of male and female students.

The estimated impacts of FFT on schooling outcomes are robust, stabilizing with the inclusion of key controls and remaining consistent as additional contextual variables are added. They are also remarkably similar across our three specifications (county-level, ED-level, and within-household). In our county-level models we find that being taught by female teachers has relatively small negative impacts on male adolescents, potentially reflecting omitted variables that led to the hiring of more female teachers and were correlated with worse outcomes for local students. Our ED and household designs allow for more granular comparisons, absorbing the effects of any local factors that impact both genders, but in these models we can only identify the relative effect of exposure to female teachers on girls versus boys.

In our third step, we turn to earlier cohorts of children and youth observed in matched 1930–1940 Census records. We evaluate the impact of FFT on educational outcomes for youth aged 15–18 in 1930, as measured a decade later in the 1940 Census, when these individuals were aged 25–28. This allows us to assess the effects of female teachers on essentially completed schooling. We find that exposure to female teachers in 1930 increased 8th grade attainment—mirroring results for 15–18 year olds in 1940—and also led to higher rates of high school graduation and college attendance among women.

For females aged 10–18 in 1930, we further examine the impact of FFT on age at first marriage and childbirth, as recorded in 1940 when these women were aged 20–28. The results indicate that women taught by female teachers were more likely to delay both marriage and childbearing.

The fourth and most consequential step is an analysis of lifetime impacts of exposure to female teachers. Building on the empirical strategies developed in the earlier stages of our study, we estimate models for individuals aged 7–14 in 1940 matched to IRS tax records from 1974 and 1979 (when they were aged 41–53), the 2000 Census (at ages 67–74), and national death records. We document enduring effects of female teacher exposure on women's long-term outcomes: women taught by female teachers ex-

perienced higher educational attainment, married better-educated husbands, and experienced higher household incomes. Exposure to female teachers also increase the longevity of women relative to men.

While our paper focuses on white students, we also evaluate the effect of FFT on Black 15–18 year olds who live in the South, where students attended segregated schools. These results are reported in the Appendix. For these adolescents as well, exposure to female teachers was beneficial to female students.

Our research adds to the growing literature on teacher gender effects by presenting new evidence that the effects of female teachers for female students extend well beyond childhood academic achievement.[6] In doing so, our work also informs the wider literature on the determinants of educational success and its long-term implications for economic well-being and longevity. These findings have potential policy relevance, particularly in contexts where gender norms are deeply entrenched, and where increasing the proportion of female teachers may help close gender gaps in student outcomes.[7] Finally, we offer a methodological contribution by demonstrating how historical US Census data can be used to estimate "neighborhood models."

We proceed as follows: In Section II we discuss research strategies used to study teacher-gender effects, review the literature, and discuss the historical context of our study. In Section III we describe our data structure and lay out the research design. Section IV reports results. Section V concludes.

II. The Role of Teacher Gender

Publicly-supported schools in the US have employed both male and female teachers since their inception, but initially relied disproportionately on male teachers. In the late 19th century, school districts increasingly employed female teachers at the primary school level, in part, it seems, because women could be paid less.[8] By the early 20th century, the teaching profession was majority female in many school districts, and there were some suggestions that boys might benefit from an increase in male teachers.[9]

As an empirical matter, in the rural US circa 1940 there was substantial variation

[6]This line of inquiry contributes to broader understandings of gender norms and stereotypes, as explored in Fortin (2005, 2015), Bertrand (2011, 2020), and Fernandez (2011).

[7]See Lee et al. (2019), who argue that in Sub-Saharan Africa, a higher share of female teachers could reduce gender disparities in achievement.

[8]As Rhey Parsons (1935) notes, "Even with the meager salaries paid, young women of ability were willing to prepare themselves for the profession of teaching, because it afforded practically the only employment opportunity open to them" (page 90).

[9]For instance, Parsons (1935) cites David Snedden, Massachusetts Commissioner of Education, who argued, "If the state is willing to pay the price, a certain proportion of men teachers should be assigned to departmental positions, not primarily because they are necessarily better teachers than women, but because it is desirable to introduce, in boys' classes at any rate, the influence of masculine personality."

across public school districts in the fraction female among teachers. We are interested in understanding how this variation in teacher gender affected students.

A. Empirical Approaches to Studying Impacts of Female Teachers

A substantial body of scholarship examines the impact of teacher gender on student achievement. Much of this work is conducted at the classroom level and relies on interaction models of the form

$$y_{ij} = \alpha_0 + \alpha_1 D_i + \alpha_2 F_{ij}^T + \alpha_3 D_i F_{ij}^T + e_{ij}, \tag{1}$$

where y_{ij} denotes an outcome for student i in classroom j, D_i is a student gender indicator (equal to one if the student is female), and F_{ij}^T is a teacher gender indicator. Assuming that y is a favorable outcome (e.g., grades or test scores), this specification allows teacher effectiveness to vary by student gender. If female teachers are about as effective as male teachers for male students, but more effective for female students, then the "main effect" of teacher gender, α_2, will be close to zero, while the "interaction effect," α_3, will be positive. In practice, researchers often augment this baseline specification with additional covariates.

Inferences based on regression (1) can be compromised by non-random selection into classrooms, by students or their parents. As a result, much of the literature focuses on environments in which students are randomly or quasi-randomly assigned to classrooms.

The study by Carrell et al. (2010) of standardized grades in STEM courses at the US Air Force Academy illustrates the use of specification (1) in a setting with random assignment.[10] Across a range of specifications, estimates of $\hat{\alpha}_2$ are close to zero, while estimates of $\hat{\alpha}_3$ are positive and statistically significant, typically between 0.10σ and 0.20σ. Carrell et al. (2010) further show that academically strong female students taught by female instructors in STEM courses are more likely to enroll subsequently in advanced mathematics courses. This result is obtained from regressions of the form

$$y_{ij} = \alpha_0 + \alpha_1 D_i + \alpha_2 F_{ij} + \alpha_3 D_i F_{ij} + e_{ij}, \tag{2}$$

where F_{ij} denotes the "fraction female" among instructors a student encounters in core STEM coursework. In this case, Carrell et al. (2010) report $\hat{\alpha}_2 \approx -0.02$ and $\hat{\alpha}_3 \approx 0.19$. These estimates imply that, for a cohort taking core STEM courses entirely from female rather than male instructors, the probability that a female student enrolls in advanced mathematics increases by roughly 17 percentage points. The use of a "fraction

[10] Mansour et al. (2022) provide a longer-term follow-up study based on the same research design.

female" measure as the key independent variable in specification (2) is the basic empirical approach we adopt in our analysis comparing male and female students across counties.[11]

Naturally, unobserved characteristics may differ across classes or cohorts exposed to different fractions of female teachers. If an omitted variable affects male and female students symmetrically, omission of the variable biases the estimated main effect α_2 but not the interaction effect α_3. If omitted factors have gender-differentiated impacts, both coefficients are generally subject to bias.

To address these concerns, some studies employ a within-classroom (or within-cohort) research design, modifying specifications (1) or (2) to include classroom or cohort fixed effects:

$$y_{ij} = \alpha_0 + \alpha_1 D_i + \alpha_3 D_i F_{ij}^T + \theta_j + e_{ij}. \tag{3}$$

This approach controls flexibly for unobserved unit-level heterogeneity. Its principal drawback is that the main effect of teacher gender, α_2, is absorbed by the fixed effect. Nevertheless, these models are well suited to settings in which the interaction effect α_3 is the primary parameter of interest. We adopt this strategy in our Census ED and household models.

B. Contexts in Which Female Students Benefit from Female Teachers

As noted earlier, the effects of teacher gender appear to be context-specific. In Appendix A, we summarize 24 economics papers published since 2000 that examine the impact of teacher gender. Of these, 16 report generally positive effects of female teachers on female students, while the remaining studies typically find little or no impact.

Notably, all evaluations conducted in low- and middle-income countries report positive impacts of female teachers on female students. For example, Lee et al. (2019) find strong positive effects on reading and math scores for girls taught by female teachers across ten Francophone African countries—a region where only 22% of teachers were female. Similarly, Muralidharan and Sheth (2016) document significant benefits for female students in India, and Gong, Lu, and Song (2018) report academic gains for Chinese girls taught by female teachers, with particularly pronounced effects among students whose mothers have low levels of education.

In contrast, studies from high-income countries—including the United States, Chile, the Netherlands, Sweden, and Germany—often find little or no overall impact of teacher gender on student outcomes. In these contexts, evidence of positive effects tends to be concentrated in STEM education, where female teachers are typically underrepresented

[11] Related strategies appear elsewhere in the literature. For example, Rothstein (1995), Robst, Keil, and Russon (1998), and Price (2010) study the impact of "fraction female" among teachers at the college level, while Nixon and Robinson (1999) and Lindahl (2016) conduct similar analyses at the high school level.

and may serve as especially salient role models for female students.

Researchers investigating the social mechanisms behind the relative efficacy of female teachers for female students often emphasize the role of teachers in challenging gender-role stereotypes that can hinder girls' educational success. The influence of such role models is especially pronounced in environments where gendered stereotypes are deeply entrenched—typically settings with few visible examples of successful women.[12]

A well-known study illustrates this key idea: Beaman et al. (2009) evaluate impacts of a "reservation policy" in India that increased the number of women in elected village-level political positions, and created random variation across villages in female political leadership. The authors show that stronger female political leadership served to "weaken stereotypes about gender roles in the public and domestic spheres." In a follow-up study Beaman et al. (2012) show that local female leadership also increased career aspirations among girls, and eliminated the gender educational gap.

C. Context for Our Analysis

We study students living in rural America circa 1940. In these rural communities there were few visible pathways to economic or educational advancement for women.

At the time, the US educational system exhibited relative gender parity in certain respects: women comprised the majority of primary school teachers, and school attendance rates were similar for boys and girls. Yet broader societal structures were marked by stark gender disparities. Few women held high-profile positions in law, medicine, management, or higher education. In rural areas especially, college-educated professional women were exceedingly rare outside of the teaching profession.[13] Among rural white women aged 24–64 in 1940, only 3.3% had completed at least one year of college and were employed. Over half of these women were teachers; the remainder were largely in clerical or secretarial roles. In contrast, 7.7% of white men had some college education and were employed, with many working as physicians, lawyers, or judges.

Given this historical context and the literature on female teacher effects, it is unsurprising that we find positive impacts of female teachers on the academic success of female students. These findings provide the foundation for the novel contribution of our research: demonstrating that the influence of female teachers extends well beyond schooling, with measurable effects on women's lifetime outcomes.

Two of our empirical findings point to the operation of a role model mechanism.

[12]See de Gendre (2023) for a meta-analysis and new empirical evidence on the same-sex role model effect. Work by Carlana (2019) highlights how teachers' communication of gendered stereotypes—such as the notion that math is easier for boys or that academic excellence is more important for boys—can be particularly detrimental to female students.

[13]See Table 8 in Appendix B for a detailed breakdown of occupations among working adults with some education in rural America in 1940.

First, the beneficial effects of female teachers are more pronounced for daughters with poorly-educated parents—an expected result, given that girls in these households are especially unlikely to encounter professional female role models outside the classroom. Second, the positive impacts of female teachers are strongest in communities with limited labor market opportunities for women beyond teaching.

We also examine potential effects of female teachers on male students. One possibility is a negative impact arising from an "attention hypothesis," in which female teachers devote more effort or engagement to female students (see, e.g., Paredes, 2014, for discussion in the context of teacher value-added). Alternatively, boys may benefit from male role models, as suggested in footnote 9. While we do estimate some negative effects of female teachers on the educational attainment of male students, we do not regard these findings as definitive. In any case, the estimated impacts are modest.

Even in the absence of direct negative effects on male students, adverse indirect effects may arise through general equilibrium forces in local marriage markets. In the rural communities we study, female teachers increase female students' educational attainment, raising rates of high school completion and college entry. These gains shift the relative distribution of human capital by gender, potentially improving women's marriage market opportunities. If men and women from these communities are situated near the lower end of the matching distribution, such improvements in women's educational attainment may reallocate matches away from local men, leading to adverse equilibrium effects on their marriage outcomes.[14] We return to this idea in the discussion that follows.

III. Data Structure and Research Design

A. Data

Our key data source is the 1940 US Census, linked to records from IRS tax filings in 1974 and 1979, the 2000 and 2010 US Censuses, and the Social Security Administration's NUMIDENT database.[15] We also incorporate the *Census Tree* linkage of the 1930 and 1940 Census files (Price et al., 2023), as well as complementary historical data sources.

Our primary analyses focus on two cohorts from the 1940 Census: children aged 7–14 and youth aged 15–18. Nearly all 7–14 year olds were enrolled in school in 1940, while only about two-thirds of those aged 15–18 remained in school. The older cohort thus allows us to examine contemporaneous schooling outcomes, such as current enrollment and educational progress. We restrict our samples to children living with at least one parent, and further limit attention to families in which all present parents were US-

[14] See, e.g., Choo and Siow (2006) and Rodriguez-Gonzalez (2021).

[15] Unless otherwise noted, we use the version of these data made available to researchers by the U.S. Census Bureau on June 6, 2018.

born.[16]

We merge the 1940 sample of children aged 7–14 to IRS records from 1974 and 1979, the 2000 and 2010 Censuses, and the Social Security Administration's NUMIDENT database using Protected Identification Keys (PIKs), restricting our analysis to cases with unique PIK matches. In addition, we conduct important supporting analyses—discussed in detail below—using female children and youth aged 10–18 from the full 1930 Census, linked forward to the 1940 Census.

Table 1 presents summary statistics for our key 1940 cohorts. The first three columns focus on youth aged 15–18. Column (1) reports statistics for the full sample, while Column (2) restricts the sample to individuals who meet the following criteria: they live with at least one parent, reside in small towns or rural areas, have not moved in the past five years, and are located in communities where teacher data can be constructed. Column (3) further narrows the sample to households with at least one daughter and one son in the target age range—a sub-sample used in our household-level models. Comparing Columns (1) and (2), we observe that youth in rural areas and small towns are slightly less likely to be female (reflecting the tendency of young women to leave home earlier than young men), are more likely to live on farms, and have parents with slightly lower levels of education. A comparison of Columns (2) and (3) shows that individuals in the "household sample" are broadly similar to other rural families.

Columns (4) through (7) in Table 1 provide analogous statistics for children aged 7–14. Column (4) includes all individuals in this age group, while Column (5) applies the same sample restrictions used for older youth. As with the 15–18 year olds, children in the target sample are more likely to live on farms and have less-educated parents. Column (6) reports statistics for individuals in the target sample who have a unique PIK, which enables linkage to 1974 and 1979 IRS tax records and to the 1-in-6 long-form sample of the 2000 Census. The characteristics of the PIK sample closely resemble those of the broader sample. Columns (7) and (8) show that matched individuals are similar to those in the larger sample summarized in Column (6).

B. Measuring the Fraction of Female Teachers with Census Data

The 1940 Census was conducted by enumerators who visited households in person, typically recording entries sequentially as they moved residence-by-residence through

[16] As discussed in Card, Domnisoru, and Taylor (2022), the IPUMS version of the data includes a hierarchy of links between children and potential parents in the household. We use the "unambiguous" link, which likely excludes a small number of children who were in fact living with parents in multi-generational households. In 1940, nearly all children aged 7–14 remained in the household, but modest numbers of 15–18 year olds (particularly females) had left home, which introduces potential selection bias in the older cohort.

their assigned enumeration districts (EDs).[17] Household and individual information was recorded on worksheets, each containing data for up to 40 individuals. Page identifiers for these worksheets were preserved during the transcription of the full-count data and are available to researchers. Individuals listed on the same page almost always lived in close proximity, and those on adjacent pages typically resided nearby as well. Thus, page number data provides a useful proxy for geographic closeness.[18] This structure enables analysis of local contextual factors, as in Logan and Parman (2017), who study neighborhood segregation, and Tan (2022), who examines social networks. We use information from nearby households to construct neighborhood-level contextual variables, such as the average characteristics of nearby adults.

Crucially, for each child in our sample, we can measure the gender composition of their potential teachers. Our key variable, fraction female of teachers (FFT_i) for child i, is derived from teachers identified in the same ED, using census occupational codes and restricting to employed teachers aged 19–70. We also record other characteristics of nearby teachers, including age, earnings, and education. We constructed alternative measures of FFT_i using the closest teachers based on worksheet page and line numbers (e.g., the nearest eight teachers), and found that results were very similar. We address concerns about measurement error in FFT_i in subsequent sections.

C. Research Design

We estimate models that build on the interaction model, regression (2),

$$y_i = \delta_0 + \delta_1 D_i + \delta_2 F_{d(i)} + \delta_3 D_i F_{d(i)} + e_i, \tag{4}$$

where y_i is an outcome variable (e.g., educational attainment), D_i is an indicator for the child's gender, $F_{d(i)}$ represents FFT among teachers in the child's ED. As discussed below, we augment this specification with vectors of child, parent, and ecological characteristics, and our baseline regressions include county fixed effects. This specification follows a differences-in-differences logic, comparing gender gaps in outcomes between high-FFT and low-FFT locations within the same county.

There are two primary threats to the validity of our approach. First, even if the assignment of female teacher share (FFT) across EDs were effectively random, families may sort strategically into neighborhoods based on teacher composition. Second, the within-county variation in FFT may itself be non-random, reflecting underlying differences in

[17] Rural EDs were designed to include no more than 1,500 people and 250 farms. In the rural areas used in our analysis, the average ED population is 1,053, compared to an overall average of 2,332. For reference, modern census tracts typically target populations of around 4,000.

[18] An exception to this pattern occurs with occasional "supplementary pages."

community characteristics that correlate with student outcomes.

The first threat—strategic sorting by families—can clearly undermine causal inference. For instance, suppose parents prefer male teachers for their sons and female teachers for their daughters, but only well-educated or higher-income families act on these preferences by relocating. In such a case, we might observe that boys perform better in low-FFT areas and girls in high-FFT areas, even in the absence of any causal effect of teacher gender. Importantly, this type of sorting would leave observable traces in the data: in low-FFT areas, boys' parents would tend to be more educated than girls' parents, while the reverse would hold in high-FFT areas.

To check for evidence of such sorting, Table 2 compares household characteristics across high-FFT and low-FFT EDs for our primary study sample—children aged 7–14 who match to 2000 Census records. Columns (2) and (3) report statistics for male and female children, respectively, in EDs with below-median FFT, while Columns (4) and (5) present comparable statistics for children in EDs with above-median FFT. Some differences are evident: families in low-FFT areas are less rural, with fewer living on farms and more residing in counties that contain a large city. However, these patterns are similar for sons and daughters. The difference-in-differences estimates in Column (8) are close to zero and statistically insignificant. This holds for key parental characteristics, including mother's and father's education and single-mother household status.

Panel B of Table 2 extends this analysis by examining whether sons and daughters are deferentially sorted into neighborhoods based on characteristics of nearby households. Again, the difference-in-differences checks reveal no systematic sorting: differences in neighborhood characteristics between low-FFT and high-FFT areas are similar for male and female children.

Finally, Panel C documents considerable variation in FFT across EDs: the average of FFT is 0.5 in below-median areas and 0.9 in above-median areas. Importantly, we see no evidence of differential sorting by child gender into high- or low-FFT locations.

The second threat is that the distribution of female teacher share (FFT) may not be "as good as random" across otherwise comparable EDs within counties. Several local factors could drive systematic variation in FFT. For example: (1) some schools may hire exclusively female teachers due to lower wage costs; (2) others may aim for a "balanced" gender mix, targeting FFT near the county average; and (3) some schools may intentionally hire at least one male teacher, perhaps based on the belief that male teachers are particularly beneficial for boys. These types of local decision-making would cause the ED-level distribution of FFT to deviate from the binomial distribution expected under random assignment.

To assess the plausibility of an "as good as random" assumption, we examine the variation in teacher gender composition across EDs within counties that have similar overall FFT. As described in the Appendix, in the first step of our analysis we bin EDs into deciles according to FFT. And then within each bin we compare the the actual and predicted distributions of the share of male teachers in EDs with 1–10 teachers. Figure 1 illustrates, focusing on the 5,301 EDs located in counties where FFT falls between 0.589 and 0.643 (which happens to be the 2nd decile). The predicted distribution assumes that the share of male teachers varies with the number of teachers in the ED, but is otherwise random, i.e., generated by a binomial process. The observed distribution closely matches the prediction: a chi-square test comparing the two yields a p-value of 0.99. Additional analyses in Appendix B confirm that this pattern holds broadly across rural counties.

We are somewhat surprised to find that the gender distribution of teachers across enumeration districts is close to random, in the statistical sense described above. This result is reassuring, but it does not fully eliminate concerns about potential bias. Specifically, it remains possible that communities hiring more or fewer female teachers differ systematically in terms of latent cultural characteristics, such as traditional or patriarchal gender norms, that could influence both the local FFT and the relative educational outcomes of girls and boys.

We consider two plausible scenarios along these lines. First, in communities where gender norms discourage broader professional opportunities for women, teaching may be viewed as one of the few socially acceptable occupations for women. In such places, schools might hire more female teachers, even as broader norms suppress girls' educational aspirations. This scenario would bias our estimates downward, making them conservative, and therefore does not concern us greatly. Second, and more troubling, is the possibility that in communities with strongly traditional views—where "a woman's place is in the home"—schools may hire fewer female teachers *and* girls may face higher barriers to educational success. We revisit this issue in Section IV.D, where we present evidence that this mechanism is unlikely in our context. Specifically, we show that average FFT is similar in communities with low female labor force participation and those with higher participation, suggesting that hiring patterns are not strongly driven by such traditional gender norms.

IV. Empirical Implementation and Results

We estimate three regression models—a county-level model, an ED-level model, and a household-level model—to assess the impact of exposure to female primary-school teachers on school enrollment and grade completion during the transition from elementary to high school, focusing on youth aged 15 to 18. These models are first estimated using

data for 15–18 year-olds in the 1940 Census, and we complement this analysis with analogous models estimated using youth aged 15–18 in the 1930 Census. These analyses provide the foundation for our subsequent investigation into whether the educational gains experienced by female students translate into improved well-being over the life course.

A. Omitted Variables and Measurement Error in an Interacted Model

As we have emphasized, our key explanatory variable, FFT, is measured with error. In addition, our regression framework may be subject to omitted-variable bias. To clarify how these issues operate in our setting, we begin by establishing a simple proposition concerning the joint consequences of omitted variables and measurement error when an explanatory variable enters both as a main effect and through an interaction with a subgroup indicator.

Proposition. Suppose the true model is an interacted specification with omitted variables,

$$y_i = \beta_0 + \beta_1 D_i + \beta_2 F^*_{d(i)} + \beta_3 D_i F^*_{d(i)} + \phi_i, \tag{5}$$

where the omitted component satisfies

$$E[\phi_i \mid D_i, F^*_{d(i)}] = \tau_0 + \tau_1 D_i + \tau_2 F^*_{d(i)} + \tau_3 D_i F^*_{d(i)}. \tag{6}$$

The researcher observes only a noisy measure of $F_{d(i)}$ of the true variable $F^*_{d(i)}$. In general the relationship between the observed and true measures can be different for male ($D_i = 0$) and female ($D_i = 1$) students. Thus, consider the group-specific linear projections:

$$F^*_{d(i)} = \theta_0 + \lambda_0 F_{d(i)} + \eta_{0i},\ D_i = 0, \tag{7}$$

$$F^*_{d(i)} = \theta_1 + \lambda_1 F_{d(i)} + \eta_{1i},\ D_i = 1. \tag{8}$$

In the "classical measurement error" case where $F_{d(i)} = F^*_{d(i)} + v_{d(i)}$ and $v_{d(i)}$ is orthogonal to $F^*_{d(i)}$, and the variances of $F^*_{d(i)}$ and $v_{d(i)}$ do not depend on the sex of the student, $\lambda_0 = \lambda_1 = \text{var}(F^*_{d(i)})/[\text{var}(F^*_{d(i)}) + \text{var}(v_{d(i)})]$. The researcher estimates a model for the outcome that includes the observed measure $F_{d(i)}$, the indicator for the group (D_i), and their interaction:

$$y_i = \delta_0 + \delta_1 D_i + \delta_2 F_{d(i)} + \delta_3 D_i F_{d(i)} + e_i.$$

Now impose the following assumptions:

A1) $\tau_3 = 0$ (the conditional expectation of ϕ_i has the same slope in $F^*_{d(i)}$ across groups).

A2) $\lambda_0 = \lambda_1 = \lambda$ (the projection of $F^*_{d(i)}$ on $F_{d(i)}$ has a common slope across groups).

Then,

$$\operatorname{plim} \hat{\delta}_2 = \lambda(\beta_2 + \tau_2) \qquad \text{and} \qquad \operatorname{plim} \hat{\delta}_3 = \lambda\beta_3. \tag{9}$$

Thus, the estimated interaction coefficient is an attenuated version of the true interaction effect. By contrast, the estimated main effect is both attenuated and contaminated by omitted-variable bias through τ_2.[19]

The key insight of the proposition extends directly to settings in which additional covariates Z_i (and their interactions $D_i Z_i$) belong in the true model. In that case, the expressions analogous to (6), (7), and (8) include these same regressors, and the derivations proceed exactly as in the simpler case. In particular, the presence of correctly specified covariates does not alter the attenuation result for the FFT interaction term, nor the role of remaining omitted variables in the FFT main effect.[20]

The proposition is helpful in interpreting our estimates. First, it provides a structured way to assess the likely direction of omitted-variable bias. In our context, economically distressed communities are more likely than others to hire female teachers, because women could be paid less than men, and more generally would invest less in schooling. Children in these same communities would also be expected to have lower levels of educational attainment, and this disadvantage would plausibly apply similarly to boys and girls. In terms of the omitted component, such considerations suggest $\tau_2 < 0$ and $\tau_3 \approx 0$. Consequently, the estimated main effect $\hat{\delta}_2$ should be biased downward, while the estimated interaction effect $\hat{\delta}_3$ is likely much less affected by omitted variables. Second, the proposition shows that measurement error generates standard attenuation bias in both coefficients. Importantly, as described below, we find that the measurement-error process in FFT is similar across male and female youth (so A2 holds), and we recover the attenuation factor λ using auxiliary measures of FFT that are independent of our Census-based proxies (from Wisconsin administrative records), allowing us to assess the extent of attenuation bias.

B. Regression Models

[19] The proof of the proposition is as follows: For $D_i = 0$, the true model (5) and conditional expectation of ϕ_i (6) imply $y_i = (\beta_0 + \tau_0) + (\beta_2 + \tau_2)F^*_{d(i)} + e_i$. Substituting the group–0 projection, $F^*_{d(i)} = \theta_0 + \lambda_0 F_{d(i)} + \eta_{0i}$, gives the slope $b_0 = (\beta_2 + \tau_2)\lambda_0$ in a linear equation that specifies y_i as a function of $F_{d(i)}$. For $D_i = 1$, the analogous steps yield $y_i = (\beta_0 + \beta_1 + \tau_0) + (\beta_2 + \beta_3 + \tau_2 + \tau_3)F^*_{d(i)} + e_i$, and substituting the group–1 projection, $F^*_{d(i)} = \theta_1 + \lambda_1 F_{d(i)} + \eta_{1i}$, gives $b_1 = (\beta_2 + \beta_3 + \tau_2 + \tau_3)\lambda_1$. In the estimation regression, $y_i = \delta_0 + \delta_1 D_i + \delta_2 F_{d(i)} + \delta_3 D_i F_{d(i)} + e_i$, we thus have $\delta_2 = b_0$ and $\delta_3 = b_1 - b_0$. Imposing A1 ($\tau_3 = 0$) and A2 ($\lambda_0 = \lambda_1 = \lambda$) yields $\delta_2 = \lambda(\beta_2 + \tau_2)$ and $\delta_3 = \lambda\beta_3$, establishing the stated probability limits.

[20] When extending our result to models with covariates, we need the standard assumption that the residual component of ϕ_i and the projection errors are assumed mean-independent of the regressors.

County Model. Our first estimating equation is the interacted regression (9), augmented with county fixed effects and additional controls:

$$y_i = \delta_0 + \delta_1 D_i + \delta_2 F_{d(i)} + \delta_3 D_i F_{d(i)} + \delta_4 Z_i + \delta_5 D_i Z_i + \omega_{c(i)} + v_i. \tag{10}$$

When estimating this regression we use the Census-based measure of FFT, which is measured with error. The inclusion of county fixed effects is intended to mitigate omitted-variable concerns by focusing on differences in the effect of FFT on boys and girls *within counties*. The remaining controls account for individual-, household-, neighborhood-, and regional-level characteristics. Specifically, we include indicators for the child's age and imputed Hispanic origin (based on surname), as well as household characteristics such as parental education and age, number of siblings, and farm-residence status. We also incorporate neighborhood contextual factors, including average parental education and income, maternal employment rates, and average employment rates for young adults (ages 19–24) by gender.[21] In addition, we control for nearby teacher characteristics that may relate to teacher effectiveness, such as average wage, age, and years of schooling. Finally, we include county-level indicators of local economic conditions. All of these controls are fully interacted with child gender. This feature is central to our design: in the interaction space, FFT is not privileged relative to other covariates.

The magnitude of the omitted-variable bias term τ_2 will tend to diminish in absolute value as additional controls are added to vector Z_i, since these covariates absorb components of the omitted factors (especially factors related to local prosperity). Under the maintained assumptions A1 and A2 in our proposition, we therefore expect the estimate of the main effect $\hat{\delta}_2$ to move toward zero as the control vector expands, while the estimate of the interaction effect $\hat{\delta}_3$ should remain comparatively stable. This is precisely the pattern we observe in the estimates reported below.

Enumeration District Model. Even with extensive controls, unobserved highly localized contextual factors—correlated with FFT and potentially affecting girls' outcomes relative to boys'—may remain. A natural way to address this concern is to include ED-level fixed effects.

Because FFT is measured at the ED level, ED fixed effects fully absorb the main

[21] The motivation for including neighborhood characteristics follows Altonji and Mansfield (2018), who show that in settings where households sort into neighborhoods based on unobserved traits, the average observed characteristics of neighbors can serve as a control function. We measure such average characteristics using 25 observations above and below each child on the census roster. Table 9 in Appendix B lists all control variables and their sources. Note that county-level variables are absorbed by county fixed effects unless interacted with child gender.

effect of FFT. Thus we estimate

$$y_i = \delta_0 + \delta_1 D_i + \delta_3 D_i F_{d(i)} + \delta_4 Z_i + \delta_5 D_i Z_i + \omega_{d(i)} + v_i, \tag{11}$$

where $\omega_{d(i)}$ denotes the ED fixed effect. As in our county model, we include all ecological controls interacted with child gender, consistent with our broader empirical strategy.

Household Model. Our final specification exploits within-family comparisons by focusing on the relative outcomes of sisters and brothers. This design mitigates concerns about parental sorting into neighborhoods, since siblings share the same household location and broader environment. Here our model includes a family-level unobserved component:

$$y_i = \beta_0 + \delta_1 D_i + \delta_3 D_i F_{g(i)} + \delta_4 Z_i + \delta_5 D_i Z_i + \omega_{g(i)} + v_i, \tag{12}$$

where child i belongs to family $g(i)$. As before, we substitute the observed ED-level FFT for its true value and include ecological controls, each interacted with child gender.

For all three models, we derive error-corrected estimates of FFT effects, using Wisconsin administrative data to recover the attenuation factor.

C. Estimates of FFT Effects on Contemporaneous Outcomes

We begin by estimating county-level regressions, which allow us to estimate both the main effect and the interaction effect of FFT on schooling outcomes during the critical transition from elementary to high school—specifically for students aged 15 to 18. Outcome measures are designed to align with our conceptual framework, which focuses on how exposure to female teachers in grades 1 through 8 influences the likelihood that students complete elementary school and continue their education. For ease of interpretation, we report regression coefficients multiplied by 100, so that each coefficient reflects the percentage point change in the probability of the outcome associated with a one-unit change in the independent variable.

Our first dependent variable is on track for 9th grade, defined as 100 if the individual has completed at least 9 years of schooling, or is currently enrolled and has completed 7 or 8 years, and 0 otherwise. This outcome captures whether students are progressing through school at a pace consistent with entering high school. In the first row of Table 3, Panel A, the main effect estimate is shown in column (2), and the interaction effect in column (3). Interpreting these estimates at face value—abstracting from potential omitted-variable bias in the main effect and without adjusting for attenuation bias—we find that being taught exclusively by female teachers (rather than male teachers) reduces the probability of being on track for 9th grade among boys by 1.1 percentage points,

while increasing the probability for girls by $3.4 - 1.1 = 2.3$ percentage points.

How important are the control variables to our estimates? Consistent with the logic of the proposition above, they substantially affect the estimated main effect but have little influence on the estimated interaction effect. Low-FFT locations were generally more prosperous than high-FFT locations, reflecting the fact that male teachers were typically paid more than female teachers. Many of our control variables capture aspects of local prosperity, so omitting them induces downward bias in the estimated main effect of FFT (separate from any attenuation bias arising from measurement error). By contrast, this same source of bias need not apply to the FFT interaction effect.

To illustrate, we estimate our model beginning with only basic child-level controls and then progressively add ecological controls (see Figure 2). As expected, in the specification without ecological controls, the estimated main effect is strongly negative. As additional controls are introduced, this estimate moves substantially toward zero. Indeed, with even richer ecological controls, the estimate might move closer still (i.e., the main effect may remain downward biased even in our preferred specification). In contrast, the estimated interaction effect is quite stable across specifications, shifting from 3.8 to 3.5.[22]

We next estimate our ED model, which includes the highly-localized ED fixed effects, and household model (which includes household fixed effects). From columns (3), (4) and (6) of Table 3, we observe that estimates of the interaction effect are remarkably similar across our three models.

If the FFT effect on girls is primarily a role model effect, we would expect the impact of FFT to be larger for girls from disadvantaged families. With this in mind, we estimate our ED and household models, restricting the sample to families in which neither parent has attained eighth grade. We find that indeed the interaction effect is larger for this sample. See columns (5) and (7).[23]

We also try a second educational attainment variable, "on track for 8th grade," which indicates that the child has 8 or more years of schooling, or is currently enrolled and has completed 6 or 7 years. Results are qualitatively similar to the 9th grade attainment variable. We also find that being taught by female teachers increases the likelihood that female students remain enrolled in school, has little effect on working, and reduces the

[22]Table 10 in Appendix B reports the coefficients on control variables. Several are notable. For example, the coefficient on "Hispanic" is -3.3 and that on "Hispanic$\times D$" is -5.4, indicating that Hispanic youth lagged in educational attainment, especially daughters. Children living with a single parent also had lower attainment: the coefficient on "only mother present" is -6.2 and on "only father present" is -5.6. Interestingly, the interaction "mother present$\times D$" is positive (3.2), while "father present$\times D$" is negative (-0.5), implying that the effect of single parenthood varies with both the gender of the parent and the gender of the child.

[23]This same result appears in the county model, but we do not report those coefficient estimates to conserve space.

rate of inactivity ("not enrolled or working").

Finally, we estimate our models for Black 15–18 year olds in rural areas of Southern states, where segregation meant that Black students were educated almost exclusively by Black teachers. We do not feature these results in our paper, because data limitations prevent us from evaluating long-run outcomes for this population. However, we do report our findings in Appendix C. As with white children, being taught by female teachers improves the educational prospects of girls relative to boys.

D. Issues Concerning the Measurement of FFT

We have concerns about three distinct measurement issues in our key independent variable, fraction of female teachers (FFT).

One concern is that in large schools, e.g., schools with multiple sections of the same grade, average school FFT need not align closely to the child's FFT if the child sorts selectively into classes. This is not a major problem in our setting for the simple reason that elementary schools in the rural areas we study are very small. In the Wisconsin data we use in our measurement-error study, we find that many students attend schools with a single teachers, and the average number of teachers in "graded schools" (which have two or more teachers) is only 2.7.[24]

A second concern is temporal coverage. Ideally, we would measure exposure to female teachers across all years of a child's primary schooling, rather than relying on a single snapshot taken during adolescence. Fortunately, for most of our target population (81.5%), we can use matched 1930 and 1940 Census records to construct enumeration-district (ED) level FFT both in 1930, when children were aged 5–8 and just beginning school, and in 1940, when they were aged 15–18. The correlation between these two FFT measures is 0.40, which is relatively high given that each is itself measured with error.

Given these two measures, we compare estimates from models that use 1940 FFT only, 1930 FFT only, the average of 1930 and 1940 FFT, or both measures simultaneously. Table 11 in Appendix B presents results from these alternative specifications, using "on track for 9th grade" as the dependent variable. The estimated interaction effects are consistent with a model in which average FFT exposure is the conceptually correct explanatory variable: each measure has a similar effect when used alone, and when both are included simultaneously, their coefficients are nearly equal, with their sum closely matching the effect obtained using the average. In the first row of Table 3, Panel B, we therefore report estimates from a model that uses the average of 1930 FFT and 1940

[24] As another example, in Maryland outside Baltimore, about 15% of students attended elementary schools with one or two teachers. The others attended "graded schools" that had 7.1 teachers on average (and this statistic likely overstates the average for students in the rural areas we study, because it includes schools in smaller urban centers). See Maryland State Board of Education (1939).

FFT to construct our key independent variables.[25]

The third, and most consequential, issue is classical measurement error. In our study we never observe the true value of FFT, denoted F^*. Instead, for the entire study area—rural America—we observe a Census-based proxy, which we now denote F^C. And as discussed in Appendix Section B.2, for much of rural Wisconsin we have a second proxy based on administrative records from the 1938/1939 *Wisconsin School Directory*, which we denote F^A. We therefore observe two noisy measures of F^*: $F^C = F^* + u_C$ and $F^A = F^* + u_A$. We assume that u_C and u_A are uncorrelated with each other, and throughout our analysis we assume that u_C is uncorrelated with any outcome variable y. Finally, we assume that this measurement-error structure is the same in Wisconsin as in the rest of the country.

Under these assumptions, Appendix B shows that a simple error-adjustment procedure can be used.[26] Specifically, we regress F^C (along with the usual covariates) on F^A. The OLS slope coefficient from this regression is a consistent estimator of the attenuation factor, λ. We estimate $\hat{\lambda} = 0.244$ (SE 0.058). This estimate can then be used to recover consistent estimates of both the main effect and the interaction effect, as shown in the Proposition above. The last row of Table 3 reports these error-corrected estimates for models with "on track for 9th grade" as the dependent variable. These estimates are roughly four times as large as their OLS counterparts reported in the first row of Table 3.

A back-of-the-envelope calculation helps connect these findings to the regressions that use the average of 1930 and 1940 FFT as the independent variable. Our analysis of Wisconsin administrative records implies

$$\hat{\lambda} \approx \frac{\text{var}(F^*)}{\text{var}(F^*) + \text{var}(u_C)} \approx 0.25,$$

abstracting from the role of additional covariates in the estimation of λ. This relationship implies $\text{var}(u_C) \approx 3 \times \text{var}(F^*)$. Now suppose, as a simplifying but admittedly strong assumption, that the Census-based measurement error process is the same in 1930 and 1940: $F^C_{30} = F^* + u_{30}$ and $F^C_{40} = F^* + u_{40}$, with $\text{var}(u_{30}) = \text{var}(u_{40})$, and that u_{30} and u_{40} are independent. Then the variance of the measurement error in the average FFT measure is

$$\text{var}\big(\tfrac{1}{2}(u_{30} + u_{40})\big) = \tfrac{1}{2}\text{var}(u_C).$$

[25] When we turn to the long-run analysis—based on children aged 7–14 in 1940—we rely solely on the 1940 FFT measure. In this setting, the 1940 FFT variable effectively captures *average* exposure to female teachers, since these children span essentially all primary-school grades (1 through 8).

[26] Our approach builds on the idea that two noisy measures of the same variable can be used to estimate attenuation bias, as in Ashenfelter and Krueger (1994). Our method, suggested to us by Lance Lochner, is a variant of this strategy.

In this case, the expected attenuation factor when using the average of 1930 and 1940 FFT is

$$\hat{\lambda}_2 \approx \frac{\text{var}(F^*)}{\text{var}(F^*) + \frac{1}{2}\text{var}(u_C)} \approx 0.40.$$

The ratio of these attenuation factors, $0.40/0.25 = 1.60$, gives the predicted relative attenuation of OLS coefficients when using averaged versus single-year FFT. This prediction aligns closely our estimates: in Table 3 the ratio of interaction estimates is $5.38/3.36 = 1.75$ in the county model, $5.34/3.39 = 1.58$ in the ED model, and $5.81/3.61 = 1.61$ in the household model.

E. A Key Threat to Validity

As discussed in Section III.C above, the primary threat to the validity of our design is the possibility that, within counties, enumeration districts (EDs) with more "traditional" paternalistic or patriarchal norms may systematically devalue girls' education and discourage maternal employment—including, plausibly, the hiring of female teachers.

To investigate this concern, we divide our data into quintiles based on the ED-level employment rate of non-teaching mothers—a measure of maternal labor force participation that is not mechanically linked to the FFT. Panel A of Table 4 shows substantial variation in maternal employment across quintiles: the rate is near zero in the lowest quintile (plausibly the most "traditional" EDs) and exceeds 22% in the highest. However, as shown in the second row, mean FFT is roughly constant across quintiles, providing no evidence that more traditional communities systematically hired fewer female teachers. Moreover, when we estimate our model separately by quintile, we find significant positive impacts of female teachers on female students in every group (Panel B).

In columns (6), (7), and (8), we further disaggregate the data by household type—rural farm households, rural non-farm households, and households in small towns (populations 1,000–2,500). Across all three settings, we again find significant positive effects of female teachers on female students.

A comparison of Panels B and C reinforces an earlier finding: the impact of female teachers is strongest for girls from families with less-educated parents. In addition, these effects are larger—across both family types—in areas where mothers have lower rates of labor force participation. Together, these patterns are consistent with a role-model mechanism that is especially salient in environments with fewer working female exemplars.

F. Additional Analyses Using Linked 1930 and 1940 Census Data

Our analysis of youth in 1940 shows that exposure to female teachers improves girls' educational prospects relative to boys, particularly in terms of being on track to tran-

sition from elementary to high school. This raises three natural questions: Do these results generalize to other cohorts? Does FFT affect completed schooling, such as high school graduation? And are there other tangible effects of FFT on young women's lives as they move from adolescence into adulthood?

To address these questions, we turn to two overlapping cohorts of youth enumerated in the 1930 Census who can be linked to responses in the 1940 Census.[27] First, we use 15–18 year olds, for whom we can estimate models similar to those we used in our 1940 sample. Second, we use a broader cohort of 10–18 year olds in 1930. For both groups we assign a measure of FFT based on the share of female teachers in their 1930 enumeration district.

For the 15–18 year olds in 1930, we measure educational attainment in 1940, when they are aged 25–28 and have essentially completed their schooling.[28] As with the 1940 cohort, we find that female teachers positively affect girls' educational attainment relative to boys. Panel A of Table 5 shows that when the dependent variable is average years of schooling, the estimated main effect is -0.12 and the interaction effect is 0.45. We continue to caution that the main effect may be downward biased due to omitted variables. In any event, the evidence suggests that any negative impact on boys is small, while the positive impact on girls is substantial.

More broadly, Table 5 reveals substantial benefits of female teacher exposure for girls, relative to boys, across multiple educational outcomes—including 8th grade completion, high school graduation, and college attendance. Although we do not perform a formal measurement-error correction for these results, there is strong reason to believe that FFT is measured with considerable error. Based on our earlier correction exercise, we expect that the true effects are substantially larger.

Given that female teachers increase the educational attainment of their female students, it is natural to ask whether they also influence related life outcomes. We begin by examining women's age at marriage and childbearing, both of which are closely linked to education (see, for example, Cummins 2025, who provides historical evidence from England and reviews the related literature). Using a sample of women aged 10–18 in 1930 (and thus aged 20–28 in 1940), we estimate a series of nine regressions (using the ED design)—one for each age cohort. The dependent variables, measured in 1940, are binary indicators for having ever married and for having at least one child. Each regression includes FFT, measured in 1930, along with the full set of family and contextual

[27] In the *Census Tree* crosswalk, match rates are quite good. For example, match rates for those those aged 15–18 year in 1930 to 1940 records are 81.5% overall: 84.6% for males and 78.3% for females.

[28] Educational status was not recorded in the 1930 Census, so it cannot be used as a dependent variable in these models.

controls used in our other 1930 models (Table 5).

Figure 3, which reports coefficients estimated using our ED specification, shows that FFT has a statistically significant negative effect on the probability of marriage for women aged 20–25, but that this effect diminishes and becomes statistically insignificant at ages 26–28. These results suggest that exposure to female teachers delays marriage but does not reduce its overall incidence, at least as measured by age 28. Consistent with patterns observed in societies without widespread access to modern contraception, age at first birth closely tracks age at marriage. Thus, exposure to female teachers also delays childbearing.[29]

We next examine whether exposure to female teachers affects labor market outcomes. For this analysis, we focus on individuals aged 15–18 in 1930 (aged 25–28 in 1940). As shown in Table 5 Panel B, we find that FFT increases labor force participation among women relative to men and raises the occupation score (occscore) for women who report an occupation.[30] Moreover, among married individuals, FFT is associated with higher occupational score for women's spouses, but not men's spouses. This suggests that women taught by female teachers fare better in the marriage market, potentially marrying more-educated or higher-earning partners.[31]

G. Lifetime FFT Effects

We complete our analysis by examining long-run effects of FFT for the sample of children aged 7–14 in the 1940 Census who meet our other study criteria, and who are

[29]The literature documents similar mechanisms operating in the educational environment. For example, Black et al. (2008) show that an increase in the compulsory schooling age in Norway during the 1960s increased women's educational attainment *and* delayed age at first birth. An analysis of this reform by Monstad et al. (2008) provides an IV estimate of the effect of education on "birth by age 20" of −0.080 (SE 0.039); the corresponding OLS estimate is −0.032 (SE 0.001).

Our setting allows a rough comparison. Consider the estimated effect of FFT on having had a birth by age 20 (denoted B) shown in Figure 3: $\Delta B/\Delta\text{FFT} = -0.020$ (SE 0.006). For this same cohort, the OLS effect of FFT on years of completed education at age 20 (denoted E) of $\Delta E/\Delta\text{FFT} = 0.359$ (SE 0.042). If the effect of FFT on B in our context were only through its impact on education, we would have

$$\frac{\Delta B}{\Delta\text{FFT}} \approx \frac{\Delta B}{\Delta E} \times \frac{\Delta E}{\Delta\text{FFT}},$$

implying $\Delta B/\Delta E \approx -0.054$. The corresponding OLS estimate of the effect of education on birth by age 20 is −0.032 (SE 0.0005).

A similar calculation for marriage by age 20 (denoted M) yields an implied effect of education of $\Delta M/\Delta E \approx -0.072$ if the FFT effect operates solely through educational attainment, compared with an OLS estimate of −0.035 (SE 0.0005).

[30]The occscore reflects the socioeconomic status or skill level associated with an individual's occupation. It assigns the median earnings for each occupation, capturing a component of permanent earnings potential that is more stable than annual wages.

[31]We also estimated models with wage as the dependent variable. While FFT has little effect on women's own wages, it does have modest positive effects on their spouses' wages—consistent with the interpretation that female teacher exposure improves women's prospects of marrying higher-earning husbands.

matched to 1974 and 1979 IRS records and to 2000 Census records (at which time they are aged 67–74). For these analyses we do not use the household model because of severe sample size limitations—a consequence of using the long-form Census records that give match rates that are at most 1-in-6 for individuals but 1-in-36 for sibling pairs. Of course, our research sample includes only those who are assigned a PIK. Fortunately, as demonstrated in Table 2, the PIK sample is reasonably similar to the broader sample.[32]

Estimated effects of FFT on lifetime educational attainment, as reported in the 2000 Census, are in Table 6. From county-level models, we infer that teacher gender had little effect on men's educational attainment (years of schooling, high school completion, or college attendance). For women, we document persistence in the positive impacts of FFT on education: being taught by female teachers in primary school moves female students up the educational ladder, increasing high school completion and college attendance. We find similar interaction effects in the ED-level models. Of course, the reported coefficient estimates are attenuated; error-corrected effects are about four times the estimates presented in the table. Thus for example, being taught exclusively by female teachers (rather than male) in childhood increased college attendance by women, relative to men, by about 10 percentage points, and had a similar impact of college attendance.[33]

When interpreting estimated impacts of FFT on schooling, in Tables 3, 4, 5, and 6, we note that results pertain for different cohorts—those born approximately 1912–1915 (Table 5), 1922–1925 (Tables 3 and 4), and 1926–1933 (Table 6). Substantial differences in reported school attainment are to be expected, given that educational attainment was rising rapidly during this era (see, e.g., Card and Lemieux, 1980), and also given that retrospective reporting of education in 2000 may be prone to exaggeration. Also, FFT impacts plausibly vary for these cohorts. Still, estimates shown in these tables tell a clear and consistent story: female teachers substantially improve the educational success of female students, especially those with low-SES families, while having at most a small negative effect on the educational success of male students.

We next turn to the impact of FFT on household income. Results in Panel A of Table 7 show that being taught by female teachers increases female students' lifetime

[32]As an additional check, we re-estimate models in Table 4 using only the subset of the data for which we can assign a PIK (rather than the 100% public-use files). Comparing the two sets of estimates provides an indication of whether the PIK process is somehow related to local FFT. As shown in Appendix Table 12, the coefficient estimates are similar for the two samples, suggesting that selection into the PIK process does not bias the estimates much.

[33]For these cohorts, lifetime educational attainment for men (but not women) was affected by the G.I. Bill (Bound and Turner 2002). We see no obviously reason this should bias estimated effects of FFT on educational attainment, but as a check we estimated regressions analogous to those reported in Table 6, but for men, and with veterans' status as the dependent variable. We find that local FFT has no impact on veteran status.

household income, relative to male students. The first rows show regressions in which the dependent variable is the "log of adjusted gross income" (log AGI) as reported in 1974 or 1979 tax filings. In these analyses the estimated impact of FFT on the men's household income (the main effect) is very close to 0, while the relative effect on women's household income (the interaction effect) is a little less than 1 percentage point, in both the county model and ED model. After inflating to adjust for attenuation bias (i.e., multiplying by four), we have a meaningful interaction effect: exposure to FFT = 1 rather than FFT = 0 increases household income for women, relative to men, by about 3 percentage points.[34]

For a smaller sample (approximately 1-in-6 individuals) we match to 2000 long-form Census records. At these ages, most individuals in our sample are retired, and reported social security income is a rough proxy of lifetime income.[35] After adjusting for attenuation bias, interaction effect estimates in these models are about 10 percent, and are higher yet for those with less-educated parents.

The positive impact of female teacher exposure (FFT) on women's lifetime household income is unsurprising. As we have shown, women taught by female teachers attained higher levels of education, which plausibly translated into improved labor market outcomes. In addition, these women appear to have matched more favorably in the marriage market.[36] To explore the labor market channel, we estimate models using the log of personal income as the dependent variable. In the county-level specification, the estimated main effect is −0.001 (standard error: 0.011), and the interaction effect is 0.049 (0.031). While the main effect is essentially zero, the interaction effect is positive, as expected, though not statistically significant—perhaps unsurprising given that we are examining income among individuals aged 67–74. To examine the marriage market channel, we estimate models analogous to those in Table 7, restricting the sample to married women and using measures of *husband's* education as the dependent variable. The results indicate that women taught by female teachers tend to marry better-educated men. Across several specifications, the estimated FFT coefficients are: 0.128 (0.042) for "husband's

[34]It is possible that these are underestimates for two reasons. First, the 1970s represent a low point in labor market inequality in the US, i.e., a time during which returns to human capital were particularly low, so lifetime returns to educational increments will be underestimated if we use annual income from that decade. Second, individuals with sufficiently low taxable income generally need not file tax returns, and this results in a standard form of selectivity bias that attenuates the estimated impact of a factor that increases education.

[35]Social security benefits are tied to lifetime earnings. The program's benefit formula compresses differences—replacing a higher share of earnings for low-income workers than for high-income workers—which might lead our estimates to understate the true impact of FFT on lifetime earnings.

[36]Nearly all women in our 2000 sample (approximately 97%) were married, either currently or previously (i.e., are widowed). We imagine that any advantages to women in the marriage market were on the intensive margin, e.g, in terms of the earnings prospects of their husbands.

completed education," 1.46 (0.66) for "husband graduated high school," and 1.12 (0.058) for "husband graduated college." The latter effect is particularly notable given that the error-corrected estimate is about 4 percentage points, and given that only 18.6% of husbands in the sample had completed college.

Substantial rural-to-urban migration occurred among the cohorts we study, especially among the well-educated (Johnson and Taylor, 2019). By 2000, approximately 65% of the women in our sample were living in urban areas. Women educated by female teachers were slightly more likely to reside in urban areas as adults and less likely to be married to men who were born on farms.[37]

Finally, we examine the impact of teacher gender on longevity. Given that exposure to female teachers increases educational attainment for girls, and that education is strongly associated with improved health and longer life expectancy, we expect that FFT may also positively affect female students' longevity.[38] In addition, female teachers may influence health-related behaviors and habits more directly.

Estimated effects of FFT on longevity are reported in Panel B of Table 7. Our first model examines survival to the year 2000. We use the full sample of 7–14 year olds in 1940 for whom we have a PIK, and define survival as appearing in either the full-count 2000 Census or in death records from 2000 onward (through 2019). Approximately 72% of individuals in these cohorts survived to 2000 under this definition. In both the county and ED models, we estimate positive and meaningful interaction effects.

After correcting for measurement error, the interaction effect suggests that being taught by female teachers exclusively (rather than male teachers) increased survival to 2000 by approximately 2 to 2.5 percentage points for women relative to men. The interaction effect appears larger yet for women from less-educated families. We replicate this analysis for survival to 2010, when individuals would be aged 77–84. Here, we find a statistically significant interaction effect only for women with less-educated parents. In our final specification, we use age at death as the dependent variable. Again, we find positive and statistically significant interaction effects in both the county and ED models. The measurement-error corrected interaction effect is approximately 0.7 years for the full sample and about one year for women born to less-educated parents.[39] Increased

[37] In regressions estimated for married women, the coefficient (standard error) on FFT is 1.45 (0.55) for the outcome "woman lives in an urban area in 2000," and −1.91 (0.87) for "husband born on a farm." These estimates are based on a 2023 version of the Census dataset, which includes minor updates relative to the 2018 version used elsewhere in our analysis.

[38] Many papers study the relationship between education and longevity, including Lleras-Muney (2022). Lochner's (2011) evaluation of "non-production benefits of education" includes a detailed discussion of mechanisms whereby education can improve health.

[39] Lleras-Muney, Price, and Yue (2022) report that an additional year of education is associated with an increase in longevity of about 0.4 years (for individuals who survive to age 35). In our context, FFT

longevity is a striking long-term benefit of exposure to female teachers.

Table 7 reports substantively negative main effects of female teacher exposure (FFT) in the longevity regressions. As discussed earlier, these estimates are likely downward biased (thanks to omitted variable bias), and it may be that these effects are closer to zero. Nonetheless, the magnitude of the negative main effects (when contrasted with the positive interaction effects) is notably larger than what we observed in the regressions examining FFT's impact on educational attainment. The interpretation of these findings remains speculative. It is possible that, in terms of health-related behaviors, boys may have benefited from greater exposure to male teachers. Another possibility is that men educated in high-FFT rural environments were disadvantaged in the marriage market, which could indirectly contribute to reduced longevity.

V. Conclusion

Our study demonstrates that female teachers in rural America circa 1940 had an enduring impact on the lives of their female students. Female teachers significantly increased girls' educational attainment. These effects were especially pronounced for girls from low-SES families and in areas with limited labor market opportunities for women—contexts where a role model mechanism is likely strongest. The benefits extended well beyond schooling: women taught by female teachers experienced delayed marriage and childbearing, and higher household income. Women taught by female teachers also experience increased longevity relative to men.

A novel element of our research is evidence that female teachers improved the marriage market outcomes of their female students. We find that women exposed to female teachers tended to marry better-educated and higher-earning spouses, suggesting that early educational influences can shape not only individual outcomes but also patterns of assortative mating. These shifts in educational and marital outcomes may have broader equilibrium effects. In communities where female educational attainment rises relative to male attainment, the improved prospects for women in the marriage market may diminish prospects for men—especially those who are less educated or economically disadvantaged. This dynamic, while not the primary focus of our study, raises interesting questions about gendered spillovers and warrants further investigation.

= 1 (versus 0) increases average education for women (relative to men) by about 0.4 years and increases longevity (again relative to men) by about 0.7 years—suggesting that the longevity gains exceed what would be expected from education alone.

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Table 1: Characteristics of White Children and Adolescents in the 1940 Census

	Aged 15–18	Aged 15–18 and meet inclusion criteria*		Aged 7–14	Aged 7–14 and meet inclusion criteria*			
	All	All	Household sample	All	All	Has PIK	Has PIK, filed tax	Has PIK, in 2000 Census
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fraction living with parent(s)	0.887	1.000	1.000	0.967	1.000	1.000	1.000	1.000
Fraction with mother only	0.110	0.094	0.085	0.083	0.066	0.056	0.054	0.049
Fraction female	0.499	0.469	0.498	0.492	0.490	0.475	0.479	0.514
Fraction enrolled at Census	0.652	0.651	0.628	0.935	0.923	0.939	0.944	0.947
Mean grade completed	9.136	8.849	8.688	4.238	4.131	4.154	4.186	4.104
Fraction in small place	0.551	1.000	1.000	0.567	1.000	1.000	1.000	1.000
Fraction living on farm	0.313	0.571	0.599	0.313	0.550	0.521	0.520	0.551
Fraction non-mover	0.750	1.000	1.000	0.765	1.000	1.000	1.000	1.000
Fraction with teacher data	0.906	1.000	1.000	0.902	1.000	1.000	1.000	1.000
Assigned PIK	–	–	–	–	–	1.000	1.000	1.000
Mean mother's education	8.407	7.861	7.680	8.650	8.101	8.342	8.420	8.505
Mean father's education	8.070	7.315	7.119	8.244	7.460	7.702	7.780	7.833
Observations**	6,714,346	2,326,299	534,995	12,971,012	4,940,564	3,000,000	2,729,000	403,000

Notes: Columns (1) through (5) are calculated using the 1940 IPUMS. Columns (6), (7), and (8) use internal Census data. *Inclusion criteria: individuals who are non-movers, living with parent(s), in rural areas, with teacher data. **Approximate sample sizes provided for columns (6), (7) and (8). Column (7) indicates the individual filed taxes in 1974 and/or 1979.

Table 2: Characteristics of 1940 Census Enumeration Districts (EDs)

		EDs with FFT < median		EDs with FFT > median		Mean differences, above − below med		Diff in diff
	All	Males	Females	Males	Females	Males	Females	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Household characteristics								
Mother's education	8.50	8.50	8.45	8.57	8.49	0.06	0.04	-0.03
						(0.03)	(0.03)	(0.04)
Father's education	7.83	7.91	7.87	7.82	7.73	-0.08	-0.13	-0.05
						(0.03)	(0.03)	(0.04)
Only mother present in household	0.05	0.05	0.05	0.04	0.05	-0.01	-0.01	0.00
						(0.001)	(0.001)	(0.001)
Living on farm	0.55	0.50	0.48	0.62	0.59	0.12	0.11	-0.01
						(0.01)	(0.01)	(0.01)
County contains a larger city	0.22	0.24	0.24	0.21	0.21	-0.03	-0.03	0.00
						(0.01)	(0.01)	(0.01)
B. Characteristics of closest neighbors								
Mother's education	8.29	8.27	8.25	8.33	8.29	0.06	0.04	-0.02
						(0.03)	(0.03)	(0.04)
Father's education	7.69	7.75	7.75	7.64	7.61	-0.11	-0.13	-0.02
						(0.03)	(0.03)	(0.04)
Living on farm	0.55	0.50	0.48	0.62	0.59	0.12	0.12	0.00
						(0.01)	(0.01)	(0.01)
C. Fraction female of teachers (FFT) in the ED								
FFT in ED	0.72	0.51	0.51	0.92	0.91	0.40	0.40	0.00
						(0.002)	(0.002)	(0.003)
Observations*	405,800	95,800	102,500	101,100	106,300	–	–	–

Notes: Statistics constructed using internal Census files. *Sample sizes are approximate. Column (8) reports the difference between columns (6) and (7). Standard errors for differences and differences-in-differences, in parentheses, are clustered by county.

Table 3: Effect of Local FFT on Educational Attainment and Activities of Adolescents Aged 15–18 in 1940

		County FE model		ED FE model, interaction effects		Household FE model, interaction effects	
	Mean of dep. var.	Main effect	Interaction effect	Overall sample	Less-educ. parents	Overall sample	Less-Educ. parents
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Baseline Regression Estimates							
On track for 9th grade	75.40	-1.07	3.36	3.39	4.33	3.61	4.69
		(0.27)	(0.29)	(0.26)	(0.34)	(0.43)	(0.54)
On track for 8th grade	85.82	-0.44	1.54	1.51	1.90	1.73	2.35
		(0.22)	(0.24)	(0.21)	(0.29)	(0.36)	(0.48)
Enrolled at Census date	65.08	-1.07	3.36	3.42	4.12	3.53	4.59
		(0.29)	(0.30)	(0.27)	(0.34)	(0.48)	(0.57)
Working at Census date	18.82	-0.36	-0.54	-0.48	-0.75	-0.21	-0.58
		(0.34)	(0.39)	(0.34)	(0.43)	(0.56)	(0.67)
Not enrolled or working	19.18	1.01	-2.50	-2.57	-3.11	-2.86	-3.57
		(0.28)	(0.43)	(0.36)	(0.46)	(0.60)	(0.74)
Counties/EDs		3,047		61,669	60,016	3,025	3,007
Observations		2,290,628		2,290,628	1,477,385	523,575	351,161
B. Alternative Regression Estimates with Dependent Variable, "On track for 9th Grade"							
Use average of 1930 FFT and 1940 FFT	75.40	-1.28	5.87	5.34	6.74	5.81	7.44
		(0.42)	(0.44)	(0.34)	(0.46)	(0.65)	(0.82)
Counties/EDs		3,046		60,481	58,658	3,020	2,993
Observations		1,901,798		1,901,798	1,197,705	422,792	275,905
Use 1940 FFT, correct for measurement error	75.40	-4.39	13.77	13.89	17.75	14.80	19.22
		(1.52)	(3.48)	(3.47)	(4.44)	(3.93)	(5.08)

Notes: Panel A presents estimated coefficients for models fit to the dependent variable in the row heading, using data for white children age 15–18 in 1940 IPUMS. In all cases the dependent variable is a binary variable equal to 100 or 0, and means reported in column (1) are for the county-model sample. Models reported in columns (2) and (3) include family controls, family controls interacted with female, neighbor controls, neighbor controls interacted with female, county fixed effects, and county-level variables interacted with female. Models reported in columns (4) and (5) contain these same controls plus fixed effects for Enumeration Districts (ED). Models reported in columns (6) and (7) contain the same controls but also household fixed effects. Models in columns (5) and (7) are fit to the subset of families in which maximum parental education is less than 8 years. Panel B provides estimates from alternative versions the regression with the dependent variable, "on track for 9th grade." Standard errors clustered by county or ED. Standard errors in the measurement-error-corrected model are calculated using the delta method (see Appendix B).

Table 4: Estimates from the ED-Level Model Estimated for Sample Subsets

	ED Quintiles, by Employment of Non-Teachers Mothers					Rural and Farm Status		
	Q1	Q2	Q3	Q4	Q5	Rural Farm	Rural Non-Farm	Small Town
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Subset Sample Characteristics								
Mother employment	0.001	0.026	0.057	0.100	0.222	–	–	–
Mean FFT	0.730	0.696	0.705	0.707	0.720	0.723	0.694	0.705
% On track for 9th grade	72.67	70.21	74.71	78.18	81.32	70.19	80.89	87.87
B. Estimates of the Interaction Effect in ED-Level Models, All Households								
On track for 9th grade	3.74	3.47	3.21	3.04	3.07	3.98	2.02	2.06
	(0.53)	(0.58)	(0.58)	(0.58)	(0.59)	(0.33)	(0.42)	(1.07)
n	456,580	463,688	456,680	458,300	455,315	1,302,944	791,352	196,332
C. Estimates of the Interaction Effect in ED-Level Models, Less-Educated Parents Only								
On track for 9th grade	4.73	4.03	3.92	4.31	4.24	4.76	2.75	3.67
	(0.69)	(0.74)	(0.76)	(0.78)	(0.87)	(0.42)	(0.62)	(1.76)
n	311,545	330,473	301,641	280,545	253,145	910,345	469,409	97,631

Notes: Calculated using the 1940 IPUMS. "Mother employment" is the rate of employment for non-teacher mothers in the ED.

Table 5: Effect of Local FFT on Educational Attainment, Measured in the 1940 Census, for Adolescents Aged 15–18 in 1930

		County FE model		ED FE model, interaction effects		Household FE model, interaction effects	
	Mean of dep. var.	Main effect	Interaction effect	Overall sample	Low-SES parents	Overall sample	Low-SES parents
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Regression Estimates for Educational Outcomes Measured in 1940 (at Ages 25–28)							
Years of schooling	9.96	-0.12	0.45	0.44	0.45	0.38	0.43
		(0.02)	(0.03)	(0.03)	(0.03)	(0.04)	(0.05)
8th grade attainment	81.9	-0.14	2.63	2.39	2.64	2.11	2.83
		(0.29)	(0.35)	(0.34)	(0.41)	(0.71)	(0.82)
High school graduation	18.8	-2.50	7.93	7.72	7.61	6.62	7.28
		(0.33)	(0.46)	(0.42)	(0.47)	(0.78)	(0.67)
Some college	12.8	-1.63	3.92	3.92	3.88	3.51	3.97
		(0.19)	(0.30)	(0.29)	(0.20)	(0.56)	(0.59)
Four years college	4.99	-0.51	1.30	1.34	0.97	1.07	1.01
		(0.12)	(0.17)	(0.17)	(0.17)	(0.35)	(0.35)
Observations		1,290,633		1,290,633	937,947	189,907	143,474
B. Regression Estimates for Other Outcomes Measured in 1940 (at Ages 25–28)							
In labor force	71.5	-1.11	4.88	4.83	4.73	4.39	4.20
		(0.18)	(0.46)	(0.40)	(0.44)	(0.80)	(0.85)
Observations		1,313,801		1,313,801	954,761	195,431	147,689
Occscore	20.9	0.12	0.51	0.48	0.63	0.76	1.09
		(0.06)	(0.13)	(0.13)	(0.16)	(0.34)	(0.39)
Observations		993,650		993,650	713,667	75,362	53,138
Spousal occscore	22.4	-0.01	0.54	0.54	0.49	0.01	-0.08
		(0.15)	(0.17)	(0.19)	(0.23)	(0.71)	(0.82)
Observations		393,849		393,849	287,202	14,810	10,953

Notes: Estimated using linked 1930 and 1940 US Census data. Table contains estimated coefficients for models fit to the dependent variable in the row heading, using data for white children age 15–18 in 1930 IPUMS. In all cases the dependent variable is a binary variable equal to 100 or 0, and means reported in column (1) are for the county-model sample. Models include control variables as indicated in Table 3. Standard errors clustered by county or ED.

Table 6: Estimated Effects of Local FFT on Educational Attainment, Measured in 2000 Census, for Children Aged 7–14 in 1940

		County FE model		ED FE model interaction effects	
	Mean of dep. var.	Main effect	Interaction effect	Overall sample	Less-educ. parents
	(1)	(2)	(3)	(4)	(5)
Years of schooling	12.08	0.021	0.108	0.125	0.130
		(0.034)	(0.039)	(0.046)	(0.070)
Completed high school	75.18	1.40	3.80	1.27	2.42
		(0.51)	(0.64)	(0.76)	(1.18)
Some college	33.11	-0.48	2.41	2.95	1.91
		(0.52)	(0.68)	(0.79)	(1.03)
Counties/EDs*		3,000		55,000	48,000
Observations*		387,000		387,000	213,000

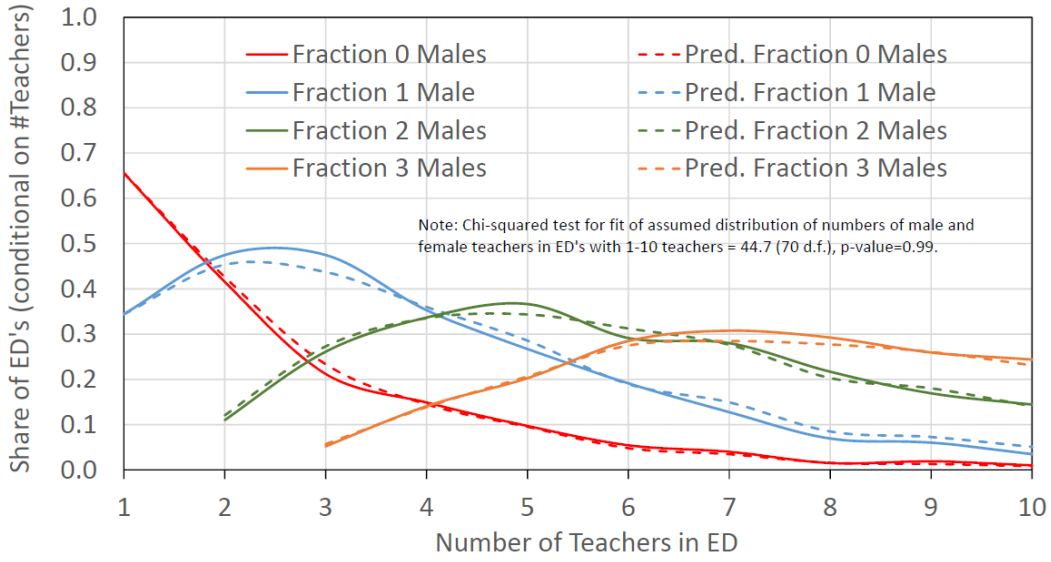
Notes: Based on data from 1940 Census internal records matched to 2000 Census internal long-form records. *Sample sizes are approximate. Means reported in column (1) are for the county-model samples. Controls are as reported in notes to Table 3. Results in column (5) are for families with maximum parental education less than 8 years. Standard errors are clustered by county or ED.

Table 7: Estimated Effects of Local FFT on Lifetime Household Income and Longevity, for Children Aged 7–14 in 1940

		County FE model		ED FE model	
	Mean of dep. var.	Main effect	Interaction effect	Overall sample	Less-educated parents
	(1)	(2)	(3)	(4)	(5)
A. Effects on Income (Measured in 1974, 1979, and 2000)					
Log Adjusted Gross Income in 1974 tax filing	9.50	-0.002	0.008	0.009	0.014
		(0.003)	(0.004)	(0.0040)	(0.005)
n		2,588,000		2,588,000	1,443,000
Log Adjusted Gross Income in 1979 tax filing	9.86	-0.003	0.009	0.009	0.013
		(0.003)	(0.004)	(0.004)	(0.006)
n		2,544,000		2,544,000	1,412,000
Log social security income in 2000 census	9.02	-0.015	0.024	0.033	0.045
		(0.004)	(0.008)	(0.009)	(0.012)
n		366,000		364,000	203,000
B. Effects on Longevity					
Alive in 2000 (in the 2000 Census or died thereafter)	72.05	-0.466	0.540	0.610	0.811
		(0.169)	(0.215)	(0.222)	(0.87)
Alive in 2010 (in the 2010 Census or died thereafter)	48.38	-0.329	0.305	0.364	0.684
		(0.178)	(0.236)	(0.244)	(0.316)
Age at death (imputed to 101 if alive in 2010)	79.55	-0.143	0.187	0.194	0.255
		(0.054)	(0.068)	(0.071)	(0.091)
n		2,997,000		2,997,000	1,699,000

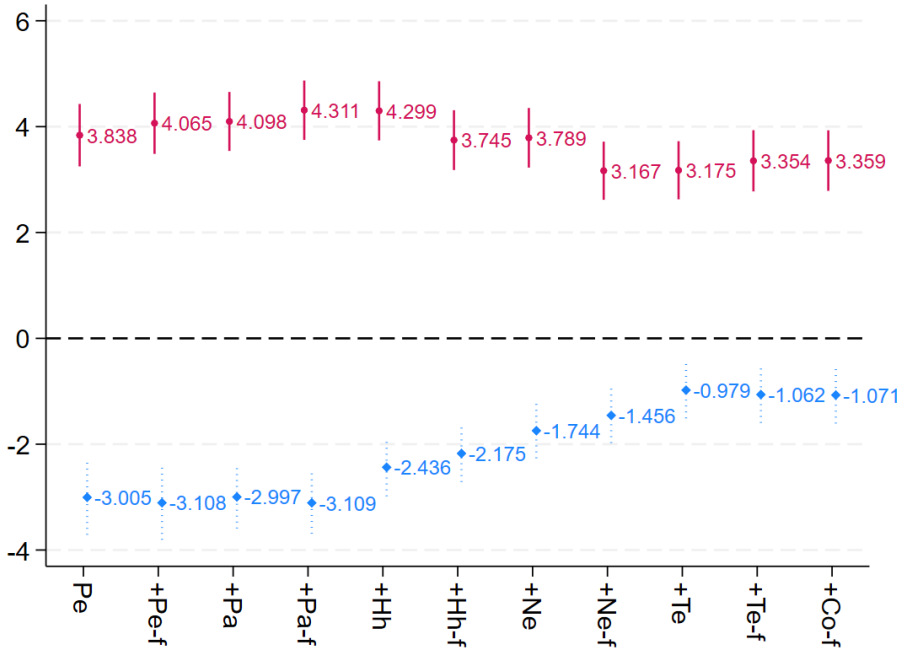
Notes: Based on 1940 US Census internal records matched to IRS records and subsequent censuses. Standard errors in parentheses. Reported coefficients are from models fit to sample/design described in column heading to income measure described in row subtitle. Models include controls as in Table 3.

Figure 1: Actual and Predicted Number of Male Teachers in EDs—Counties with ≈ 0.625 FFT



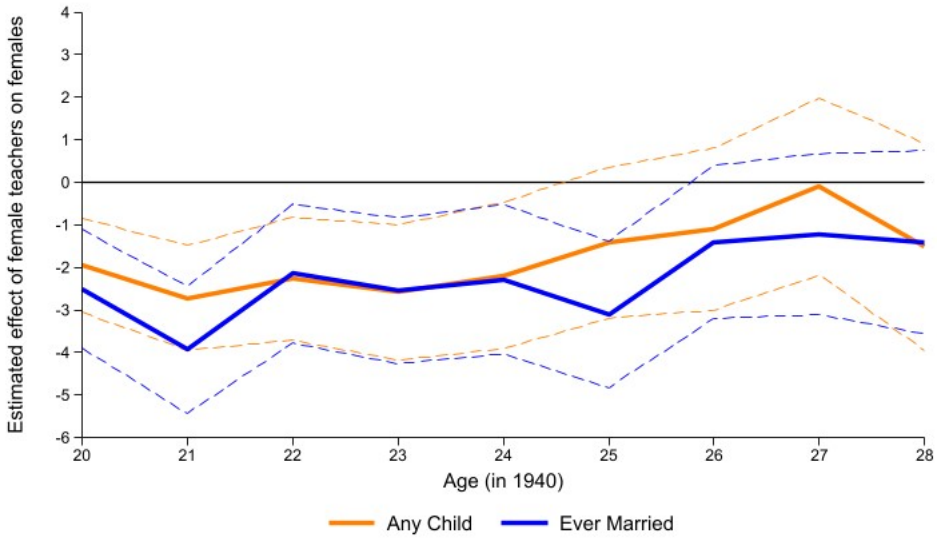
Data: 1940 PUMS. The graph is based on 5186 EDs in rural areas and small towns in 321 counties for which $0.589 < \text{FFT} < 0.643$. Predicted fractions are calculated assuming the probability of a female teacher in any ED is equal to the mean FFT among EDs in the same county with the same number of teachers.

Figure 2: Effect of Adding Controls on Key Coefficient Estimates: Analysis of 9th Grade Attainment (1940)



Point estimates and 95% confidence intervals of *main effects* are in blue; *interaction effects* are in red. On the far left of the graph, we provide estimates for a model with in which the only control variables are related to the child—age and Hispanic status (Pe). Then we add those same variables but with gender interactions (Pe-f). Then moving to the right we add parental variables (Pa); then parental variables interacted with child gender (Pa-f); then household controls (Hh); then household controls interacted with gender (Hh-f); then neighborhood controls (Ne); then neighborhood controls interacted with gender (Ne-f); then teacher controls (Te); then teacher controls interacted with gender (Te-f); and finally also county control variables interacted with gender (Co-f).

Figure 3: Estimated of Local FFT on Marriage and Childbearing, Measured in the 1940 Census, for Females Aged 10–18 in the 1930 Census



Notes: These are estimated effects of FFT on "any child" and "ever married" estimated in nine separate regression, using the ED design, by age for cohorts of females aged 10–18 in 1930 census records matched to 1940 census records. In all cases the dependent variable is a binary variable equal to 100 or 0.

APPENDICES: FOR ONLINE PUBLICATION

Appendix A Extended Literature Review

To help place our findings into intellectual context, we briefly review the recent literature in economics on the effects of teacher gender on student outcomes. We summarize findings from 24 articles, published since 2000 in economics journals, which provide empirical evidence on this matter. To keep the review manageable we ignore papers that were published earlier or in non-economics journals. Of these 24 papers:

- Six papers study students in low-income countries or study students from low-SES families in medium-income countries: children in rural India (Muralidharan et al. 2016); students in China (Gong et al. 2018, and Xu and Li 2018, and Eble and Hu 2020); students in ten African countries (Lee et al. 2019); and girls with with poorly-educated mothers in Chile (Paredes 2014). *All of these papers show that female teachers improve outcomes for female students, while having little or no adverse impacts on male students.* Several of them show also that beneficial effects are larger for girls from lower-SES backgrounds.

- Ten papers report largely beneficial impacts of female teachers for female students in medium- and high-income countries: Dee (2005), Dee (2007), Carrell et al. (2010), Winters et al. (2013), and Egalite et al. (2018), and Mansour et al. (2022) in the US; Lindahl (2016) in Sweden; Paredes (2014) in Chile; and Lim and Mear (2017 and 2020) in Korea. Many of these papers find beneficial impacts of female teachers for female students *specifically in science and math.*

- Six papers report weak evidence, mixed evidence, or no evidence of beneficial impacts of female teachers for female students—all in high-income countries: Bettinger et al. (2005), Krieg (2005), Price (2010), and Gershenson et al. (2016) in the US; Holmlund et al. (2008) in Sweden; and Cho (2013) in 15 OECD countries.

- Only two papers feature negative impacts of female teachers for female students. Antecol et al. (2015) find that female students in math to tend not fare well when matched to female teachers in the Teach for America organization, which places volunteer teachers in low-income neighborhoods. As the authors emphasize, this effect is driven primarily by female teachers who lack a strong math background. Gørtz et al. (2018) study Danish preschools, where teachers are overwhelmingly female (91% female at the median school). In that environment there appears to be a small benefit (in terms of test scores) to having at least one male teacher in the preschool.

In addition, we include in our review three relevant papers which focus primarily on the extent to which teachers' behaviors, expectations, and attitudes affect outcomes among female and male students: Sansone (2017), Alan et al. (2018), and Carlana (2019). The latter two studies specifically evaluate gender stereotypes held by teachers, and ask how they differentially impact male and female students.

Finally, listed separately from the other articles (at the end of the list), are five related papers that focus primarily on the impacts of teacher and student gender on grading and assessment.[40] These papers find evidence of consequential grading bias among teachers—by comparing systematic gender differences in boys' and girls' non-blind versus blind student test scores. Importantly, *none of the papers show that female teachers are biased in favor of female students in terms of grading.*

We draw several lesson from this literature, relevant to our own work:

[40] Mechtenberg (2009) provides an interesting theoretical contribution.

First, female teachers seem to matter most for female students in educational situations in which those teachers are plausibly counter-stereotypical, in particular in less developed countries, where women often are marginalized in society, and female students lag male students in achievement.[41]

Second, the research on the consequences of teacher gender stereotyping (Alan et al. 2018, and Carlana, 2019) provides further direct evidence on a mechanism by which female teachers can play a key role in improving outcomes for female students without causing any harm to male students.

Third, research on gender biases in grading is important because it demonstrates a key mechanism whereby teacher/student genders can be consequential for students. One might hypothesize that female teachers would be more biased in favor of female students than male teachers—which of course might benefit female students while harming male students. However, evidence in the literature does not favor this hypothesis.

Finally, while none the research we review examines the outcomes in our study (e.g., completed education, and other lifetime outcomes), the literature does provide a sensible way of tracing out likely impacts of female teachers in environments in which those teachers are counter-stereotypical role models. Consider, e.g., the persuasive analysis of young students in India by Muralidharan et al. 2016. This study shows that each year of student exposure to a female teacher increases female academic achievement by 0.034σ, while boys' outcomes are unaffected. Thus, female students who have female teacher during their first six years of school plausibly increase achievement by more that 0.2σ, and plausibly have stronger non-cognitive skills as well (as shown in many papers reviewed below). Many of these students would thus be better positioned and motivated to progress to higher grades.[42] Hopefully, researchers will examine this important question; our work does so, though of course in a different educational context.

Papers published in economics over the past 20 years (since 2002) are listed on the next few pages. Results are reported as statistically significant only if the p value is ≤ 0.05. Papers on grading bias are featured separately at the end of the list.

[41] Even the two papers finding negative effects of female teachers on female students could be construed to be consistent with the importance of counter-stereotypical role models. In Danish preschools, studied by Gørtz et al. (2018), *male* teachers are rare, and might be viewed as counter-stereotypical. The authors note that the positive impact of male teachers seems to be largest for boys who lack male role models at home. As for work by Antecol et al. (2015), we find plausible the authors' suggestion that "... math anxiety among primary school female teachers in conjunction with female student endorsement of gender stereotypes may be leading to poorer math achievement among female students but not male students."

[42] If so, this would lower the male/female ratio of students in upper grades found in rural Andra Pradesh, which now exceeds 1 for students in most castes (see Bagde, et al. 2022).

Bettinger et al. (2005)	At public colleges in Ohio, female faculty in freshman classes are found to impact subsequent course selection and major choice by female students. Female faculty have mixed statistically significant effects on female students—increasing engagement in mathematics and statistics, geology, sociology, and journalism and communications, while reducing engagement in biology, physics, economics, and political science.
Dee (2005)	An analysis of data from the National Education Longitudinal Study of 1988 (NELS88) shows that 8th graders are perceived by opposite-gender teachers to be more inattentive, more disruptive, and more likely to "rarely complete homework." Similar results pertain also for students taught by other-race teachers.
Krieg (2005)	Being matched to teachers by gender is shown to have no effect on achievement—as measured by the Washington Assessment of Student Learning—among 3rd/4th graders in Washington.
Dee (2007)	Using data from the NELS88, the author finds that assignment to a same-gender teacher improves achievement by about 0.05σ for both boys and girls. Teacher-gender impacts are also found along other dimensions, e.g., boys are more likely to be seen as disruptive when the teacher is female, and girls are about 0.1σ less likely to report "science is not useful for their future" when their teacher is female.
Holmlund et al. (2008)	In a study of upper-secondary students in Stockholm, the authors find little evidence of teacher-student gender match effects.
Hoffmann et al. (2009)	In a study of University of Toronto students, same-gender instructor effects are found to be small—increasing grade performance by 0.05σ or less, and lowering the likelihood of dropping a class by 0.04σ.
Carrell et al. (2010)	When female students at the US Air Force Academy are assigned to female professors, grades in mandatory math and science courses increase by about 0.1σ, while having little impact on male students. For female students with high SAT scores, the effect is closer to 0.2σ. Among these latter students, female professors increases the likelihood of follow-on STEM coursework, and increases graduation with a STEM degree by nearly 0.3σ.
Price (2010)	An OLS analysis of students in first-semester STEM courses at public four-year universities in Ohio shows that each additional female STEM instructor reduces persistence in a STEM major by about one percentage point (about 0.04σ) among males after the first semester, while having no statistically significant effect for female students. (A complementary IV design yields statistically insignificant estimates.)
Winters et al. (2013)	The authors examine the effect of female teachers on standardized math and reading scores of students in Florida—grades 4 through 10. No effects of teacher gender are detected in grades 4 and 5. In grades 6 through 10, however, having a female teacher improves math achievement among female students (by 0.02σ to 0.04σ), while improving male achievement by slightly less (0.01σ to 0.03σ).

Cho (2013)	The author estimates the effects of teacher-student gender matching on achievement, measured by the Trends in International Mathematics and Science Study in 15 OECD countries. As the authors indicatet: "The results provide little support for the conjecture that students benefit from teacher-student gender matching."
Paredes (2014)	For female 8th graders in Chile, having a female teacher increase scores in math (by 0.03σ to 0.05σ), in natural sciences (by 0.02σ to 0.03σ) and in social sciences (by 0.02σ to 0.03σ). Estimated effects are larger for girls with poorly-educated mothers, and are close to 0 among girls with well-educated mothers. Teacher gender does not generally impact achievement among male students.
Antecol et al. (2015)	The authors study impacts of teachers in the Teach for America corps, who teach primary school students in disadvantaged neighborhoods. Having a female teacher does not affect male student test scores (math or reading) and does not affect female student reading test scores, but has a *negative* impact of female student math test scores (about -0.1σ). This outcome is driven primarily by female teachers who do not have a strong math background. The authors suggest that "... math anxiety among primary school female teachers in conjunction with female student endorsement of gender stereotypes may be leading to poorer math achievement among female students but not male students."
Gershenson et al. (2016)	Using data from the Education Longitudinal Study of 2002 (ELS), the authors evaluate teacher expectations of their 10th grade students. They find little effect of teacher gender matching on teacher expectations. (However, they do find that non-Black teachers have lower expectations of Black students.)
Muralidharan et al. (2016)	The authors study education in public primary schools in rural Andhra Pradesh (India). In this setting, girls and boys have similar levels of knowledge in both math and language at the start of primary school (girls indeed have a slight advantage in language), but then girls lose ground relative to boys—losing 0.01σ/year in language and 0.02σ/year in math assessments. Against this backdrop, female students taught by female teachers see gains of 0.034σ/year, relative to boys, in test scores (math and language combined). Boys suffer no adverse effects from being taught by female teachers.

Lim and Meer (2017)	Female Korean middle school students are found to have higher standardized test scores when taught by female teachers (impacts are near 0.1σ). In contrast, teacher gender has little effect on test scores of male students. When female students are taught by female teachers, they are more likely to report that they have an equal chance to participate in class (6 pp), that their teacher encourages expression (6 pp) and that the subject they are studying is their favorite subject (4 pp).
Sansone (2017)	In an analysis of data from the High School Longitudinal Study of 2009, the author finds significant effects of teacher gender on students' interest and self-confidence in math and science (e.g., female teachers raise self-confidence in female students by 0.05σ). However, effects are not significant in specifications that control also for a set of variables measuring teacher behaviors, expectations and attitudes. This leads the author to suggest, "Teacher beliefs about male and female ability in math and science—as well as how teachers treat boys and girls in the classroom—matter more than teacher's own gender."
Alan et al. (2018)	In a low income area in Istanbul, female primary students (but not their male counterparts) are found to have lower performance on math and verbal tests when taught by teachers who hold traditional views on gender roles. Given a four-year exposure to a teacher, a one standard deviation increase in the teacher gender stereotype construct has a -0.21σ effect on female student math scores and a -0.16σ effect on verbal scores.
Egalite et al. (2018)	Using data from six school districts across the US, the authors study effects of student-teacher gender matches for students, grades 4–8, in terms of student perceptions (student feels cared for by their teacher, student interest and enjoyment of classwork, quality of teacher-student communication, clarity in teaching style and methods, etc.). Student perceptions and attitudes are generally more favorable when they have a same-gender teacher: effect sizes (when significant) are in the range of 0.04σ to 0.09σ. (Similar effects pertain for student-teacher race matching.)
Gong et al. (2018)	In an analysis of data from the 2014 China Education Panel Survey (CEPS), a nationally representative survey covering middle schools, the authors find that for female students, having a female teacher improves test scores by about 0.2σ, relative to boys, and increases self-assessed learning performance by more than 0.3σ. Impacts on test scores are larger for girls whose mothers have ≤ 9 years of education (0.27σ) and ethnic minorities (0.46σ). Also, many non-cognitive outcomes are studied: for example, when taught by a female teacher, girls are less likely to report feeling depressed, blue, or unhappy at school, and report stronger social acclimation and satisfaction.

Gørtz et al. (2018)	Using administrative data, the authors study Danish children enrolled in preschool (2006–2007)—a setting in which very few teachers are male (at the median preschool, 91% of teachers are women). Here, an increase in the fraction *male* among teachers increases student test scores at age five, e.g., a 10 pp increase in fraction male increases reading test scores by 0.035σ for boys and 0.026σ for girls.
Xu and Li (2018)	In a study utilizing two waves of the CEPS, the authors study effects of teacher-student gender matches on students. The authors find that boys are little affected by teacher gender. Having a female teacher improves girls' self-reported interactions with students, and increases exam scores in math (0.15σ) and Chinese (0.10σ)
Carlana (2019)	The author measures gender stereotypes using an implicit associaton test (IAT) among teachers in Northern Italy, and examines impacts on students of exposure to teacher stereotypes. A one SD increase in a teacher's IAT score is found to decrease females' 8th-grade math performance by more than 0.03σ relative to male counterparts. Math teacher stereotypes also induce female students to self-select into less demanding academic tracks, and have a negative impact on their self-confidence in math.
Lee et al. (2019)	This paper studies 2nd and 6th grade students in ten African countries. Teacher gender has no effect on boys, but being taught by female teachers is beneficial for girls—increasing girls' 2nd grade reading and math scores by 0.1σ, and 6th grade reading by 0.1σ and math by 0.2σ.
Eble and Hu (2020)	In a study of Chinese students using data from the CEPS, the authors examine the impact of teacher gender on gender stereotypes. When girls who believe themselves to have low math ability are assigned to female math teachers, they score 0.45σ higher on math exams relative to other similar children. They are also less likely to perceive math as difficult, and less likely to aspire to traditionally-female jobs.
Lim and Meer (2020)	Using data from the Seoul Education Longitudinal Study (2010), which tracks students from grades 7–12, the authors find that female 7th graders taught by a female teacher score higher on standardized tests (by 0.19σ). These effects are persistent, resulting in increases in test scores that range from 0.16σ in grade 8 to 0.25σ in grade 12. Also, having a female 7th grade math teacher increases the likelihood female students attend a STEM-focused high school, take advanced math, and aspire to a STEM degree.
Mansour, et al. (2022).	The authors study post-graduation outcomes for students at the US Air Force Academy. Female students with more female professors are more likely to complete a masters degree in STEM and less likely to have a professional degree. Effects on male students are in the opposite direction and about half as big.

Recent Papers that Focus on Grading/Assessment Bias

Lavy (2008)	The author exploits blind and non-blind exam score data on matriculation exams taken by Israeli high school students. Bias in grading—the difference between blind and non-blind scores—on average disadvantages boys (the disadvantage is 0.05σ to 0.18σ in nine academic subjects. However, no clear associations were found between *teacher gender* and this grading bias.
Lindahl (2016)	This work evaluates impacts of teacher gender on student outcomes among 9th graders in Sweden. Female students are found to have higher math test scores when taught by female teachers; we infer that being taught by all-female teachers (rather than all-male) increases math test scores by 0.21σ. Also, there is evidence of gender bias, favoring girls, in terms of receiving a favorable teacher-assigned score on the "School Leaving Certificate." However, if anything, female teachers are less likely than male teachers to be biased in terms of such "grading up."
Puhani (2018)	The author examines the impact of teacher gender on the teachers' recommendations for middle school type in Germany, and also on the actual middle school type attended. There are "virtually no teacher gender effects;" the only exception is an increase in high-track recommendations for boys by male teachers (0.07σ).
Lavy and Sand (2018)	The authors find lasting consequences of teacher gender biases in grading on 6th grade students in Tel-Aviv. For example, they show that teacher gender bias—measured as the difference between boys' and girls' average gaps in non-blind and blind test scores—has a positive effect on subsequent enrollment in advanced high school math courses for boys relative to girls. A one SD decrease in teacher gender bias is estimated to reverse the gender gap in advanced course participation, from 3.2 pp in favor of boys to 4.6 pp in favor of girls.
Terrier (2020)	The author utilizes a data set of French students (in grades 6–11), which includes both blind and non-blind test scores, to explore teacher bias in grading. Many middle school teachers tend to bias grading in favor of female students, and these biases are found to affect educational outcomes. For example, being exposed to a math teacher with a one standard deviation bias in favor of girls, compared to one with no bias, increases by 0.10σ the likelihood a female student chooses a scientific track in high school.

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Appendix B Supplementary Material

B.1 Additional Tables and Figures

Occupations in the Rural US *circa* 1940

In the text we suggest, "Among rural white women aged 24–64 in 1940, only 3.3% had completed at least one year of college and were employed. Over half of these women were teachers; the remainder were largely in clerical or secretarial roles. In contrast, 7.7% of white men had some college education and were employed, with many working as physicians, lawyers, or judges."

For Southern Black adults in rural areas, only 1.3% of women and only 1.2% of men had some college and were employed. More than 60% of these Black women and about one third of these Black men were teachers. (These statistics are relevant for our study of impacts of FFT on Black students in Appendix B.3).

Table 8 documents the statistics cited in these paragraphs.

Table 8: Occupations of Working Adults with Some College Education, Rural US in 1940

White women		White men	
Percent who had some college and were employed	3.28	Percent who had some college and were employed	7.68
Percent in occupation:		Percent in occupation:	
Teachers	55.74	Farmers (owners and tenants)	14.47
Stenographers, typists, secretaries	5.89	Managers, officials, proprietors	14.05
Clerical and kindred workers	3.49	Teachers	10.67
Managers, officials, proprietors	3.46	Salesmen and sales clerks	5.15
Bookkeepers	3.24	Clerical and kindred workers	3.51
Salesmen, sales clerks	3.20	Clergymen	3.14
Nurses, professional	2.64	Physicians and surgeons	2.98
Farmers (owners and tenants)	1.83	Lawyers and judges	2.63
Musicians and music teachers	1.74	Bookkeepers	2.51
Operative and kindred workers	1.68	Operative and kindred workers	1.98

Black women in the South		Black men in the South	
Percent who had some college and were employed	1.30	Percent who had some college and were employed	1.23
Percent in occupation:		Percent in occupation:	
Teachers	61.94	Teachers	31.65
Private household workers	12.01	Farmers (owners and tenants)	19.68
Service workers, not private household	1.69	Clergymen	6.92
Stenographers, typists, secretaries	1.53	Laborers	6.43
Laundresses, private household	1.49	Farm laborers and wage workers	4.71
Operative and kindred workers	1.42	Professors and instructors	2.46
Nurses, professional	1.37	Managers, officials, proprietors	2.41
Professors and instructors	1.35	Mine operatives, laborers	1.82
Clerical and kindred workers	1.32	Farm laborers, family workers	1.77
Beauticians, manicurists	1.09	Private household workers	1.40

Notes: 1940 IPUMS data, individuals aged 24–64 living in rural areas and small towns (less than 1,000 inhabitants), with at least one year of college education.

The Distribution of FFT across EDs

As discussed in Section II.C, an important step in evaluating our research design is to examine the distribution of FFT across EDs within counties. Remarkably, we find that the distribution appears quite "close to random" as a particular way that we now describe.

In step 1 of our analysis, we bin EDs into deciles, based on the ED-level FFT, e.g., the 1st decile has FFT < 0.589, and the 2nd decile has $0.589 <$ FFT < 0.643. There are many EDs in each decile, e.g., the 2nd declie has 5186 EDs located in 321 counties.

In step 2, we observe that within each decile there is some modest variation in mean FFT, conditional on the number of teachers in the ED (we consider EDs with 1 to 10 teachers).[43] For example, in the 2nd decile, we observe the following:

Teachers	1	2	3	4	5	6	7	8	9	10
Mean FFT	0.656	0.653	0.616	0.617	0.625	0.603	0.618	0.594	0.617	0.619

We then ask if, conditional on the number of teachers in an ED, the distribution of male and female teachers is close to what one would predict if teacher gender assignment were random.[44] We find that, in fact, observed outcomes are often close to these predictions.

Consider, for example, evidence presented in the first panel of Figure 4 (on the next page), which shows selected predicted and actual proportions of male teachers for the 2nd decile. Here the predicted proportion of EDs with 0 male teachers is 0.095 and the actual proportion is 0.097, the predicted proportion with 1 male teacher is 0.286 and the actual proportion is 0.267, and so forth. Remarkably, for this decile we cannot reject the null hypothesis that teacher gender is randomly assigned conditional on the number of teachers.

Our table shows such graphs for three of the ten county average-FFT decile groupings—the 2nd, 5th, and 9th deciles (all of which are constructed for rural areas and small towns). The 2nd decile has 5186 EDs in 321 counties, for which $0.589 <$ FFT < 0.643; the 5th decile has 5301 EDs in 276 counties, for which $0.712 <$ FFT < 0.736; and the 9th decile has 5449 EDs in 304 counties for which $0.822 <$ FFT < 0.857. In the three illustrated cases, actual proportions are quite close to predicted outcomes. More generally, in seven of our deciles we cannot reject the null hypothesis noted in the previous paragraph.

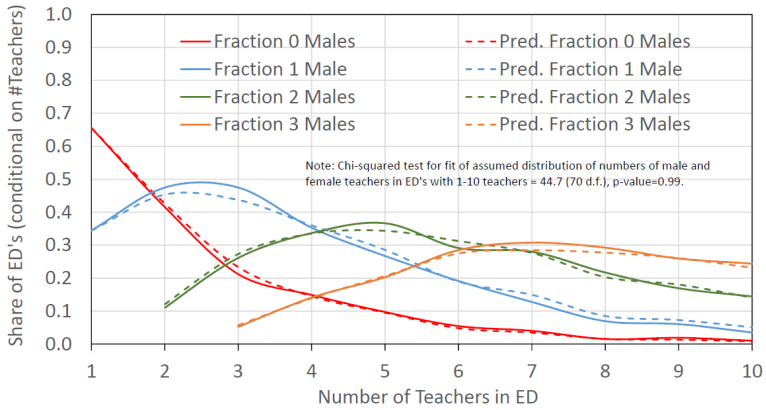
We find these results to be surprising, and somewhat comforting. However, as we emphasize in the text, threats to the validity of our research design remain.

[43] There are few EDs with more than 10 teachers in our sample of rural areas and small towns.

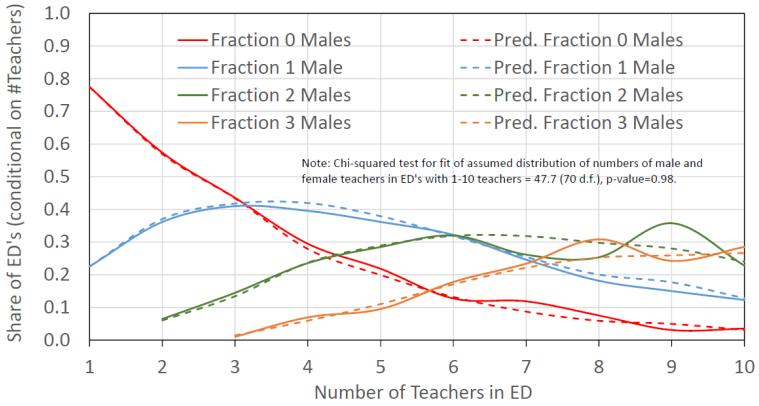
[44] This is a simple binomial distribution. In our case, in EDs with 5 teachers, mean FFT is 0.625. If teacher gender were randomly assigned for such EDs, the proportion of EDs with 0 male teachers would be $0.625^5 = 0.095$, the proportion with 1 male teacher would be $5 \times (1 - 0.625) \times 0.625^4 = 0.286$, and so on.

Figure 4: Actual and Predicted Number of Male Teachers in EDs for Three Sets of Counties—at the 2nd, 5th, and 9th Deciles of Average FFT

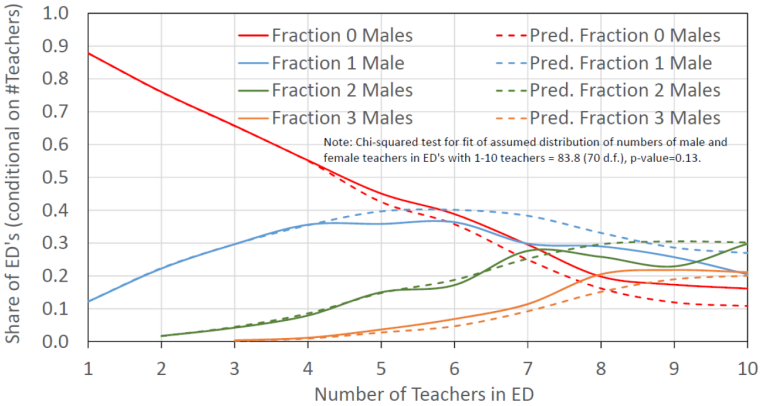
Decile 2 (FFT ≈ 0.625)



Decile 5 (FFT ≈ 0.725)



Decile 9(FFT ≈ 0.805)



Control Variables in our Regressions

Our regressions include many control variables. We provide the list here for all regressions except those that utilize the 1930–1940 Census match (i.e., all regressions except those reported in Table 5) :

- Child Characteristics. We include a gender indicator ($D = 1$ if the child is a daughter), and also age indicators (e.g., for 15, 16, 17, or 18) and a Hispanic indicator, which are also interacted with child gender.
- Parental and Household Characteristics. We have indicator variables for parental education, parental age, number of siblings, farm status, only mother present, only father present, Hispanic status, farm residence, small town residence, and residence outside a metro area. Also included is the difference between the mother's and father's education. In addition, we include interactions of these variables with child gender.
- Characteristics of Neighbors. Using the closest 50 neighboring households, we construct neighbors' average fathers' education, mothers' education, fathers' income, mothers' employment, employment rate of men aged 19–24, employment rate of women aged 19–24, percent residing on a farm, and emergency employment among fathers. Also included are interactions of these variables with child gender.
- Characteristics of Nearby Teachers. We include teacher variables that are plausibly be related to teacher effectiveness: average wage, age, and years of schooling, and interactions of these variables with child gender.
- County Variables. We include variables related to county prosperity (Table 12). Our model has county fixed effects, which absorb the main effect of these variables; we are including the child-gender interaction effects only.

Table 9 provides sources for key variables. In Table 10 we show the estimated effects of our control variables in one key regression—the regression with "on track for 9th grade" reported in the first row of Panel A of Table 3.

For regressions the utilize the 1930–1940 Census match, we are missing all variables that rely on parental education, including the parents' own education, and the average education of nearby neighbors. We also do not have the variables constructed from income and wage data, e.g., the average income of nearby neighbors. (However, because we match 1930 households to 1940 households, we can use parental education variables, as reported in 1940.) For 1930, we also use a few control variables that are not used in 1940: literacy of parents, occscore, radio ownership, and rent.

Table 9: County-Level Variables, with Sources

County-level variable	Source
Average occupational score, men aged 35–54 Average occupational score, women aged 35–54 Average number of children, women aged 21–30 Indicator, city with population 25,000+ in county Doctor/population ratio for Whites	1940 IPUMS, 100 percent sample
Minimal distance to a normal school	Author's calculations, using locations listed in *The Factors Operating in the Location of State Normal Schools* by Harry C. Humphreys (Columbia University, 1923)
Minimal distance to a 4-year college	Author's calculations, using data for institutions founded before 1940, *College Scorecard,* US Department of Education
Fraction of White births delivered in hospital Infant mortality for Whites, 1941	Data provided by Michael Haines
Population density (persons per square mile), 1940 % of rural farm dwelling units with electric lighting, 1940 % of rural farm dwelling units with running water, 1940 % of workers employed in agriculture, 1940 % of dwelling units with private bathtub or shower, 1940 Sum of major war supply contracts and projects ($)	*County and City Data Book*[a]

[a]US Bureau of the Census, *County and City Data Book,* Consolidated File: County Data, 1947–1977. Distributed by Inter-University Consortium for Political and Social Research, 2012-09-18. https://doi.org/10.3886/ICPSR07736.v2

Table 10: Estimated Coefficients for Control Variables, Regression with Dependent Variable, "On Track for 9th Grade," White 15–18 Year Olds in 1940

FFT ("main effect")	-1.071
	(0.274)
FFT × D ("interaction effect")	3.359
	(0.292)
Daughter (D)	-0.665
	(4.577)
Only mother present	-6.158
	(0.169)
Only mother present × D	3.191
	(0.181)
Only father present	-5.554
	(0.225)
Only father present × D	-0.534
	(0.296)
Hispanic	-3.031
	(1.092)
Hispanic × D	-6.020
	(0.772)
Farm	0.255
	(0.183)
Farm × D	1.285
	(0.170)
Small town	0.074
	(0.247)
Small town × D	-0.130
	(0.180)
Difference, mother's and father's education	-2.057
	(0.031)
Difference, mother's and father's education × D	0.608
	(0.024)
Outside metro area × D	0.445
	(0.338)
Neighboring fathers' education	0.699
	(0.071)
Neighboring fathers' education × D	-0.651
	(0.074)
Neighboring mothers' education	1.397
	(0.075)
Neighboring mothers' education × D	0.861
	(0.083)
Neighboring fathers' income	-0.057
	(0.027)

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Table 10 – *Continued from previous page*

Neighboring fathers' income × D	-0.041
	(0.017)
Neighboring mothers' employment	-3.833
	(1.012)
Neighboring mothers' employment × D	-1.415
	(1.003)
Employment rate, neighboring men aged 19–24	-2.991
	(0.265)
Employment rate, neighboring men aged 19–24 × D	1.561
	(0.273)
Employment rate, neighboring women aged 19–24	1.541
	(0.275)
Employment rate, neighboring women aged 19–24 × D	-1.465
	(0.307)
Wage of neighboring men aged 19–24	0.253
	(0.020)
Wage of neighboring men aged 19–24 × D	-0.160
	(0.023)
Wage of neighboring women aged 19–24	0.107
	(0.019)
Wage of neighboring women aged 19–24 × D	-0.039
	(0.023)
Percent of neighbors residing on farm	-8.539
	(0.299)
Percent of neighbors residing on farm × D	1.728
	(0.277)
Emergency employment among neighboring fathers	-1.340
	(1.009)
Emergency employment among neighboring fathers × D	-3.810
	(0.892)
Neighborhood-county imputed data	-1.244
	(0.127)
Neighborhood-county imputed data × D	0.084
	(0.145)
Teacher wages	0.093
	(0.000)
Teacher wages × D	0.130
	(0.024)
Teacher age	-0.059
	(0.012)
Teacher age × D	-0.001
	(0.011)
Teacher years of schooling	0.519
	(0.058)

Continued on next page

Table 10 – *Continued from previous page*

Teacher years of schooling × D	-0.015
	(0.059)
Occupational score of men in county × D	-0.055
	(0.088)
Occupational score of women in county × D	0.155
	(0.062)
Percent of births in hospital × D	0.862
	(0.638)
Doctor-population ratio × D	0.473
	(0.124)
Infant mortality ×100 × D	-0.941
	(0.472)
Fraction employed in agriculture × D	0.028
	(0.012)
Population density ×1000 × D	0.195
	(0.131)
Percent of households with bathtub × D	-0.070
	(0.015)
Percent of farms with water × D	0.020
	(0.009)
Percent of farms with electricity × D	-0.008
	(0.006)
War economy spending ×1000 × D	-0.002
	(0.121)
Distance to normal school ×100 × D	-0.335
	(0.150)
Distance to college ×100 × D	0.369
	(0.309)
Number of children, women aged 21–30 × D	-1.930
	(0.737)
Indicator, city in county × D	-0.609
	(0.319)
County fraction women married × D	0.491
	(0.706)
Employment rate women 34-55 × D	20.940
	(2.240)
County living index × D	-0.042
	(0.007)
County-state data imputation	-5.747
	(20.48)
County-state data imputation × D	1.516
	(1.012)
Also included:	
Parental education FE	

Continued on next page

Table 10 – *Continued from previous page*

Parental education FE $\times$ D
Parental age FE
Parental age FE $\times$ D
Child age FE
Child age FE $\times$ D
Number of siblings FE
Number of siblings FE $\times$ D

Notes: Table shows an extended list of coefficient estimates for the regressions presented in the Panel A of Table 3, columns (2) and (3).

Analysis Using 1930 and 1940 FFT Measures

Table 11 presents results from alternative specifications—using 1930 FFT and 1940 FFT—to study the impact of FFT on the outcome, "on track for 9th grade" among 15–18 year olds in 1940. We find the following: substituting 1930 FFT for 1940 FFT yields only modest changes in coefficient estimates. When both measures are included in the model, each has a statistically significant and positive interaction effect with female gender. Also, as expected, the estimated effects are larger when we use the *average* of the 1930 and 1940 FFT measures. (This result is also reported in the first row of Panel B in Table 3.) The average provides a less noisy proxy for cumulative exposure to female teachers, thereby reducing attenuation bias in both the estimated main effect and interaction effect.

Table 11: Effect of Local FFT on Being "On Track for 9th Grade" in 1940—Models with Alternative FFT Measure

		County FE model		ED FE model, interaction effects		Household FE model, interaction effects	
	Mean of dep. var.	Main effect	Interaction effect	Overall sample	Low-SES parents	Overall sample	Low-SES parents
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Baseline Models, Using FFT in 1940 (at Ages 15–18)							
1940 FFT	75.40	-1.08	3.36	3.39	4.33	3.61	4.69
		(0.27)	(0.29)	(0.26)	(0.34)	(0.43)	(0.54)
Observations		2,290,628		2,290,628	1,447,385	522,715	350,670
B. Models Using FFT in 1930 (at Ages 5–8) or Average of FFT in 1930 and 1940							
1930 FFT		-0.58	3.55	2.80	3.42	3.43	4.26
		(0.31)	(0.34)	(0.25)	(0.33)	(0.51)	(0.65)
Avg., 1930 and 1940 FFT		-1.28	5.87	5.34	6.74	5.81	7.44
		(0.42)	(0.44)	(0.34)	(0.46)	(0.65)	(0.82)
Observations		1,901,798		1,901,798	1,197,705	422,792	275,905
C. Models that Include Both FFT in 1930 and FFT in 1940							
1940 FFT	75.40	-0.93	2.99	3.13	4.17	3.08	4.11
		(0.29)	(0.30)	(0.28)	(0.38)	(0.47)	(0.61)
1930 FFT		-0.33	2.87	2.30	2.74	2.71	3.31
		(0.31)	(0.33)	(0.25)	(0.33)	(0.51)	(0.65)
Observations		1,901,798		1,901,798	1,197,705	422,792	275,905

Notes: Table contains estimated coefficients for models using the independent variable(s) in the row heading, using data for White children age 15–18 in 1940 IPUMS. In all cases the dependent variable is a binary variable equal to 100 or 0, and means reported in column (1) are for the county-model sample. Models include control variables as indicated in Table 3. Standard errors clustered by county or ED.

Analysis for the PIK Sample

One of our concerns is that individuals assigned a PIK differ from other individuals in terms of the relationship between FFT and educational outcomes. Table 12 repeats the key analysis from Table 3, but only for individuals for whom a PIK was assigned. Reassuringly, results reported in Table 12 are very similar to those reported in Table 3.

Table 12: Effect of Local FFT on Education Attainment and Activities in 1940, White Adolescents Assigned PIKs

		Estimated effect of local fraction of female teachers (FFT)					
		County FE model		ED FE model, interactions only		Household FE model, interactions only	
	Mean of dep. var.	Main effect	Interaction effect	Overall sample	Less-educ. parents	Overall sample	Less-Educ. parents
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
On track for 9th grade	78.37	-1.31	3.85	4.18	5.40	3.14	3.89
		(0.28)	(0.32)	(0.31)	(0.42)	(0.52)	(0.70)
On track for 8th grade	88.41	-0.43	1.67	1.82	2.38	1.30	1.64
		(0.21)	(0.24)	(0.24)	(0.34)	(0.42)	(0.58)
Enrolled at Census date	67.25	-1.42	3.72	4.12	4.84	3.56	3.80
		(0.30)	(0.33)	(0.33)	(0.44)	(0.60)	(0.78)
Working at Census date	19.38	-0.35	-0.61	-0.72	-0.74	-0.20	-0.42
		(0.32)	(0.38)	(0.37)	(0.47)	(0.65)	(0.81)
Not enrolled or working	16.79	1.16	-2.66	-2.88	-3.59	-2.45	-2.51
		(0.27)	(0.42)	(0.38)	(0.52)	(0.65)	(0.85)
Counties/EDs		3,000		60,000	60,000	3,000	2,900
Observations*		1,347,000		1,347,000	836,000	209,000	132,000

Notes: This table replicates Table 3 but includes only individuals assigned a PIK (using Census internal records). *Sample sizes are approximate. See notes for Table 3 for other details.

B.2 Evaluating and Correcting Attenuation Bias Using Wisconsin Administrative Data

To assess the magnitude of attenuation bias in our FFT estimates arising from measurement error, we conduct a "case study" using data from Wisconsin, for which we observe teacher gender composition from both the 1940 Census *and* independent administrative records from the 1938/1939 *Wisconsin School Directory*. These administrative records list the names of all teachers in high schools and in "graded schools" (schools with two or more teachers), permitting a near-complete enumeration of teacher gender for a large set of rural and small-town enumeration districts. Comparable administrative publications are not available for most other states.

Before using Wisconsin to study measurement error, we assess how representative the state is relative to the national sample. Across several demographic and economic indicators, Wisconsin closely resembles the rest of the country. Employment shares in agriculture are similar (40.6% vs. 35.2%), as is population density (152.9 vs. 158.7 persons per square mile). Parental schooling in Wisconsin is only modestly higher than national averages (8.14 vs. 7.85 years for mothers; 7.46 vs. 7.31 years for fathers). Teacher characteristics also align broadly with national patterns: Wisconsin has a slightly higher share of female teachers (0.76 vs. 0.71), slightly lower wages ($879 vs. $950), and slightly lower average education (13.9 vs. 14.6 years) and potential experience (ages 29.4 vs. 32.8). Occupational scores for working-age adults are very similar as well.

To confirm that the *relationships* between FFT and educational outcomes in Wisconsin resemble those elsewhere, we re-estimate our core models using Wisconsin alone. Table 13 (at the end of this appendix) reports these results. For the "on track for 9th grade" outcome, estimates are very similar in Wisconsin and the full national sample (see Table 3), suggesting that Wisconsin is broadly representative in terms of the underlying relationships of interest.[45]

We next construct an alternative FFT measure using the Wisconsin School Directory. We focus on rural enumeration districts and small towns with fewer than 2,000 inhabitants. Using school location information in the directory, we match high schools and graded schools to townships and towns listed in the Census ED descriptions. Two research assistants independently recorded the number of female teachers in each school; in nearly all cases teacher gender could be inferred from names. We retained only EDs for which both coders could confidently identify the local schools. We denote the resulting administrative measure by F_d^A.

In short, in our study we never observe the true value of FFT, F^*. Instead, for the entire study area—rural America—we have a Census-based proxy, which we denote F^C (for clarity), and for much of rural Wisconsin we have a second proxy based on administrative records, F^A. These proxies are noisy versions of F^*: $F^C = F^* + u_C$ and $F^A = F^* + u_A$. Measurement errors u_C are u_A are assumed to be uncorrelated, and throughout or analyses we have assumed u_C is uncorrelated with any outcome variable y.

Now, as demonstrated in the proposition in Section IV.A, measurement error in our setting

[45]The "on track for 8th grade" outcome differs because 8th grade completion was nearly universal in Wisconsin, leaving little cross-sectional variation. Other outcomes are somewhat noisier but broadly in line with national estimates.

generates standard attenuation bias in both the main effect and the interaction effect—just as in a univariate regression—provided that the measurement error in FFT is the same for teachers of girls and teachers of boys. This condition is very natural in our context since girls and boys attend the same schools. Under this assumption, correcting for measurement error reduces to estimating the familiar attenuation factor λ from the classical errors-in-variables model. Our estimate of λ is the slope from a regression of F^A on F^C (and additional covariates). The following section provides the justification for this approach.

Consistent Estimation Using Two Proxies for an Unobserved Regressor

We estimate our models (10), (11), and (12) using F^C rather than the true FFT, F^*. In model (10), F^C appears in both the main effect and the interaction effect, but as just noted, under our maintained assumptions error correction requires estimation of a single attenuation parameter, λ, and of course the same is true in models (11) and (12). Also, insights from the univariate case extend immediately to models with additional covariates. Thus, for clarity we present the univariate case: here the goal is simply to recover a consistent estimate of β_1 in the true model, $y = \beta_0 + \beta_1 F^* + v$.

Step 1: Regression of y on the Census Proxy F^C (Entire Study Sample)

Using the full Census data, we estimate our specified model using F^C and the full set of covariates. The univariate analog is

$$y = \delta_0 + \delta_1 F^C + \varepsilon,$$

in which case the OLS slope is

$$\hat{\delta}_1 = \frac{\text{Cov}(y, F^C)}{\text{Var}(F^C)}.$$

Step 2: Regression of One Proxy on the Other (Wisconsin Sample)

Using the Wisconsin administrative-records data, we estimate the projection of F^A on F^C, again including the full set of covariates. The univariate analog is

$$F^A = \alpha + \lambda F^C + \eta,$$

with OLS slope

$$\hat{\lambda} = \frac{\text{Cov}(F^A_{\text{WI}}, F^C_{\text{WI}})}{\text{Var}(F^C_{\text{WI}})}.$$

Step 3: Construct a Consistent Estimators of β_1

Our estimator of β_1 is the ratio of $\hat{\delta}_1$ to $\hat{\lambda}$. Consider that ratio:

$$\frac{\hat{\delta}_1}{\hat{\lambda}} = \frac{\left[\frac{\text{Cov}(y,F^C)}{\text{Var}(F^C)}\right]}{\left[\frac{\text{Cov}(F^A_{\text{WI}},F^C_{\text{WI}})}{\text{Var}(F^C_{\text{WI}})}\right]} = \left[\frac{\text{Cov}(y, F^C)}{\text{Cov}(F^A_{\text{WI}}, F^C_{\text{WI}})}\right]\left[\frac{\text{Var}(F^C_{\text{WI}})}{\text{Var}(F^C)}\right].$$

If the Wisconsin sample is representative of the national data with respect to the relevant moments of F^C, then

$$\frac{\hat{\delta}_1}{\hat{\lambda}} \approx \frac{\text{Cov}(y, F^C)}{\text{Cov}(F^A, F^C)}.$$

The ratio above has the same form as the classical two-proxy instrumental variables estimator. Under standard assumptions, the probability limit of this expression is β_1.[46] If both proxies were available for the entire study area, the IV estimator

$$\hat{\beta}_1^{IV} = \frac{\text{Cov}(y, F^C)}{\text{Cov}(F^A, F^C)},$$

would be applied to the full sample. Our three-step procedure is an adaptation of the two-proxy IV correction for attenuation bias, tailored to the case in which $\text{Cov}(F^A, F^C)$ can be estimated only from the Wisconsin subsample.

Estimate of $\hat{\lambda}$

We estimate the projection of F^A on F^C using a regression that includes the full set of covariates, including ED-level fixed effects.[47] This yields an estimated attenuation factor of $\hat{\lambda} = 0.244$ (SE 0.058), which we use to correct estimates presented in the bottom row of Table 3. Under assumptions stated in the Proposition in Section IV.A, our correction delivers consistent estimates of the interaction effect, and also of the main effect in the county model when there is no omitted-variable bias ($\tau_2 = 0$). We assume that the estimated attenuation factor applies across models. According, we assume that the measurement error structure for teachers in our subsamples—teachers of children of less-well-educated parents and siblings in household sample—is the same as in the full sample.

Standard Error Estimation

Standard errors of the corrected coefficients are calculated using the delta method. If a parameter of interest $r = f(\beta)$ is a differentiable function of an estimated parameter vector $\hat{\beta}$ with variance–covariance matrix $\hat{V}$, the delta method approximates the variance $v(\hat{r})$ by $F'(\hat{\beta})\hat{V}F(\hat{\beta})$, where $F(\beta)$ is the gradient of $f(\cdot)$. In our application, $r = \beta_1/\lambda$, and we assume that $\hat{\beta}_1$ and $\hat{\lambda}$ are uncorrelated (i.e., that there is no correlation between estimates from the measurement error model—fit to the Wisconsin data—and those from the main model). Under this assumption,

$$\hat{v}(\hat{r}) = \frac{1}{\hat{\lambda}^2}\hat{v}(\hat{\beta}_1) + \frac{\hat{\beta}_1^2}{\hat{\lambda}^4}\hat{v}(\hat{\lambda}).$$

[46] The IV logic is familiar. F^A and F^C contain the same underlying signal F^*, while their measurement errors, u_A and u_C, are independent. Consequently, F^C is correlated with the true regressor F^* (and therefore with F^A through the shared signal), but uncorrelated with the measurement error in F^A. Thus F^C satisfies the *relevance* and *exogeneity* requirements for a valid instrument for F^A.

[47] In this regression, we assume that the error structure is the same for girls' and boys' teachers, and we easily fail to reject this null hypothesis.

Table 13: Effect of Local FFT on Educational Attainment and Activities in 1940, Wisconsin

		Estimated effect of local fraction of female teachers (FFT)					
		County FE Model		ED FE model, interaction effects		Household FE model interaction effects	
	Mean of dep. var.	Main effect	Interaction effect	Overall sample	Less-educ. parents	Overall sample	Less-educ. parents
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
On track for 9th grade	72.9	1.54	3.25	3.43	4.23	3.89	3.86
		(1.49)	(1.81)	(1.72)	(2.12)	(2.35)	(2.82)
On track for 8th grade	94.4	1.60	-1.25	-1.16	-1.10	0.25	-0.65
		(1.09)	(1.26)	(0.94)	(1.25)	(1.89)	(2.48)
Enrolled at Census date	61.4	2.31	0.74	0.49	1.03	0.72	2.29
		(1.54)	(1.61)	(1.71)	(2.08)	(2.54)	(2.49)
Working at Census date	22.4	-4.17	2.07	2.07	1.29	7.32	5.96
		(1.80)	(2.29)	(2.16)	(2.57)	(3.54)	(4.07)
Not enrolled or working	19.3	0.57	-2.30	-2.10	-2.56	-7.19	-8.37
		(1.13)	(2.09)	(2.14)	(2.53)	(2.87)	(3.53)
Counties/EDs		71		1,702	1,689	71	71
Observations		67,883		67,883	47,322	16,605	11,790

Notes: Table contains estimated coefficients for models fit to the dependent variable in row headings, using data for White adolescents age 15–18 in 1940 IPUMS, for Wisconsin only. Otherwise this table reports models exactly as in Table 4.

C Estimates of FFT Effects on Educational Outcomes for Black Students

Our paper studies the effects of teacher gender composition on the educational success of white female students. Here we ask if Black female students similarly benefited from being taught by female teachers in the environment we study. We focus on the US South, where 83% of Black adolescents resided in 1940 and where segregation meant that Black students were educated almost exclusively by Black teachers.

We analyze contemporaneous outcomes, but not long-run outcomes becuase match rates for the Black population prove to be quite low and samples are too small for reliable inference.

Table 14 gives summary statistics similar to those in Table 1. Column (1) provides statistics for all Black 15–18 year olds in the US, and column (2) narrows to Southern states. We notice that parents in the South have lower levels of education than in the North. In column (3) we further narrow our sample to those who live in with parent(s) in rural ares, who have not moved in the past five years, and for whom we can construct suitable teacher data. In the case of Black adolescents, we find that 22% live in EDs with just one Black teacher, so to limit measurement error in FFT introduced by one-teacher EDs, we include only those living in EDs with two or more Black teachers.[48]

The parents of the Black adolescents we study have very low levels of education, about three years less than parents of white rural adolescents. Similarly, mean grade completion among Black adolescents is much lower than for rural white adolescents (6.1 versus 8.8 years of completed schooling). Nearly three quarters of our sample lives on a farm, and almost one in five live with a mother only. Sample characteristics of the household sample are very similar to the overall sample.

Table 15 reports analyses which parallel those for white adolescents (Table 3). In our headline model, with a dependent variable "on track for 9th grade," we estimate interaction effects that are about the same as for white youth, at least in the county and ED models. Estimated effects in the household model are much less precise than for white youth, which is expected, as the Black sample is only about one tenth the size of the white sample. The same same observations pertain when we use the dependent variable "on track for 8th grade."

Far fewer Black youth than white youth are "on track" for either 8th or 9th grade education, primarily because nearly half of Black adolescents are not enrolled in school. Thus, we try two other benchmarks of educational attainment—6th grade completion (achieved by 57% of our sample) and 8th grade completion (achieved by 28%). With these dependent variables, we again estimate large and statistically significant interaction effects. Estimated effects are similar for the full sample and for the sub-sample with less-educated parents, which is not surprising as virtually all Black parents fall in the latter category.

We have seen that for white adolescents FFT interaction effect estimates are subject to serious attenuation bias due to measurement error in FFT, and to recover unbiased estimates of interaction effects we need to multiply OLS estimates by about four. We have not conducted a comparable

[48] When we make a similar restriction for our white sample, this has no material effect on estimates.

exercise for Black FFT in the South, but we suspect that measurement error is, if anything, a larger problem for Black teachers, because there are typically fewer Black teachers than white teachers per ED. Thus, if we multiply interaction effect estimates by four we likely have conservative estimates. When we do so, the interaction effect on our benchmark regression, "being on track for 9th grade" is about 10. This effect is similar to that estimated for white students.

The estimated main effects in our regressions are larger (in absolute value) for Black youth than for white youth. Perhaps Black male students experienced modest gains from being taught by Black male teachers. Further investigation is warranted.

Overall, our findings are consistent with the literature on teacher role model effects, which shows that students tend to benefit from being taught by same-sex or same-race teachers in contexts in which there are otherwise few professional role models with whom the student identifies. This depiction accurately depicts female students we study, both white and Black. As for male students, white students would have had no shortage of professionals who might serve as exemplars; nearly all high-prestige local professionals (doctors, lawyers, professors, etc.) were white men, as were nearly all nationally prominent professionals. Black male students in the rural South, on the other hand, had little exposure to college-educated professionals of their same sex and race, other than teachers; it follows that these students might benefit from being taught by Black male teachers.[49]

We have one additional set of analyses for 15–18 year olds in Southern states—a "falsification exercise" in which we assign Black students the FFT of nearby white teachers, and white students the FFT of Black teachers, while otherwise retaining race-specific control variables. Since education in the South was segregated, the relevant FFT for students of a given race is their own-race FFT. If significant effects appear for opposite-race FFT, this indicates potential omitted variable bias (e.g., unobserved local environmental factors, not entirely race-specific, which drive both female education and the employment of women as teachers). We restrict attention to ED's that contained at least two teachers of each race, thereby avoiding comparisons of local areas that were entirely of one race or the other. Table 16 reports the estimated interaction effects from the ED design using own-race teacher FFT and the opposite-race FFT. Reassuringly, a comparison of columns (1) and (2) shows that for Black youth, only the same-race FFT affects student outcomes, while a comparison of columns (3) and (4) shows the same for white youth.

[49] Gershenson et al. (2021) provide a thoughtful overview of research on the value of same-race teachers for minority students, and provide important new empirical evidence. In an evaluation of high school students in North Carolina, they find that the high school dropout rates are lower for economically-disadvantaged Black males when they have Black teachers. They find suggestive evidence, moreover, that this beneficial impact is larger when the Black teachers are male.

Table 14: Characteristics of Black Adolescents Aged 15–18 in the 1940 Census

	Adolescents aged 15–18			
			In small towns/rural areas, non-movers with teacher data	
	All	In Southern states	All	In household sample
	(1)	(2)	(3)	(4)
Fraction living with parent(s)	0.795	0.788	1.000	1.000
Fraction with mother only	0.198	0.186	0.197	0.170
Fraction female	0.517	0.516	0.482	0.500
Fraction enrolled at Census date	0.512	0.481	0.555	0.566
Mean grade completed	6.733	6.299	6.074	6.076
Fraction in small place	0.593	0.689	1.000	1.000
Fraction living on farm	0.437	0.519	0.721	0.763
Fraction non-mover	0.768	0.815	1.000	1.000
Fraction with teacher data*	0.521	0.564	1.000	1.000
Mean mother's education	5.501	5.116	4.883	4.938
Mean father's education	4.533	4.133	3.877	3.919
Sample Size	1,011,416	839,866	190,138	58,245

Notes: Data are from the 1940 IPUMS. *Individuals are recorded as "having teacher data" only if they are in an enumeration district with with at least two Black teachers.

Table 15: Effect of Local FFT on Educational Attainment and Activities in 1940, Southern Black Adolescents

		Estimated effect of local fraction of female teachers (FFT)					
		County FE model		ED FE model, interaction effects		Household FE model, interaction effects	
	Mean of dep. var.	Main effect	Interaction effect	Overall sample	Less-educ. parents	Overall sample	Less-educ. parents
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
On track for 9th grade	42.3	-2.68	2.84	2.86	3.28	1.04	1.46
		(1.12)	(1.19)	(1.11)	(1.19)	(1.76)	(1.87)
On track for 8th grade	51.9	-2.27	2.62	2.67	3.30	2.26	3.08
		(1.21)	(1.19)	(1.14)	(1.22)	(1.80)	(1.48)
Completed 8th grade	27.5	-2.06	3.51	3.39	3.63	1.91	1.92
		(0.90)	(1.06)	(0.96)	(1.00)	(1.48)	(1.53)
Completed 6th grade	56.7	-1.96	2.88	2.95	3.06	1.95	2.02
		(1.04)	(1.14)	(1.07)	(1.14)	(1.69)	(1.80)
Enrolled at Census date	55.5	-2.79	1.90	1.82	2.34	1.46	1.75
		(1.36)	(1.26)	(1.16)	(1.22)	(1.72)	(1.80)
Working at Census date	34.2	1.25	-1.90	-1.95	-1.89	-2.32	-2.19
		(1.57)	(1.51)	(1.38)	(1.46)	(1.84)	(1.95)
Not enrolled or working	16.4	0.18	0.99	1.21	0.78	3.18	2.85
		(0.84)	(1.16)	(1.16)	(1.22)	(1.53)	(1.58)

Notes: Table contains estimated coefficients for models fit to the binary dependent variable in row headings, using data for Black children age 15–18 in 1940 IPUMS. In all cases the dependent variable is a binary variable equal to 100 or 0, and means reported in column (1) are for the county-model samples. Models include controls as reported in the notes to Table 4. Models in columns (5) and (7) are fit to subset of families in which maximum parental education is less than 8 years. Standard errors clustered by county or ED.

Table 16: "Falsification Exercise" for Southern Black and White Adolescents: Interaction Effect of FFT (ED Model) with Own-Race and Opposite-Race FFT

	Black students		White students	
	Black FFT	White FFT	Black FFT	White FFT
	(1)	(2)	(3)	(4)
On track for 9th grade	2.72	-1.25	0.79	4.16
	(1.37)	(1.62)	(0.95)	(1.29)
On track for 8th grade	2.73	1.40	0.41	3.92
	(1.37)	(1.61)	(0.87)	(1.21)
Completed 8th grade	3.65	0.20	0.02	3.20
	(1.19)	(1.46)	(0.94)	(1.31)
Completed 6th grade	3.28	1.82	-0.71	1.72
	(1.31)	(1.54)	(0.70)	(0.94)
Observations	117,118	117,118	158,724	158,724

Notes: Notes: Table contains estimated coefficients for models fit to the dependent variable in row headings, using data for Southern Black and White children age 15–18 in 1940 IPUMS. In all cases the dependent variable is a binary variable equal to 100 or 0. Control variables are as reported in the notes to Table 4. The sample is restricted to enumeration districts that have at least two White and two Black teachers, and non-missing teacher wage, age, and teacher education information.